

Simplicial Complexes

Dr. Elena Xinyi Wang

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Housekeeping

- Problem presentation: assigned at the end of each class
- Homework assignments: only written/programming
- Homework 1 , Due 01/10.
- Homework 0 .

We officially have a TA! :)

Recall from last time

Topology	A rule telling us what it means for points to be locally near one another
Open set	Every point has some wiggle room around it that is in the set $(-1, 3)$
Closed set	The set contains its limiting points $[1, 3]$
Metric topology	Distance defines “nearby”, balls give neighborhoods; unions of balls give open sets

$$\mathbb{R}, \underline{|x_2 - x_1|}$$

Recall from last time

Maps	Continuous functions (nearby inputs don't jump to far-away outputs)
Connected	The space is one piece
Homeomorphism	Two spaces are the same topological shape. You can continuously transform one into the other and continuously undo it.
Embedding	Put one space inside another without changing its topology. No identifying/gluing different points together, and the topology on the image must agree with the original one.
Homotopy	A continuous movie of maps: g gradually changes into h , the parameter t is time.

map.

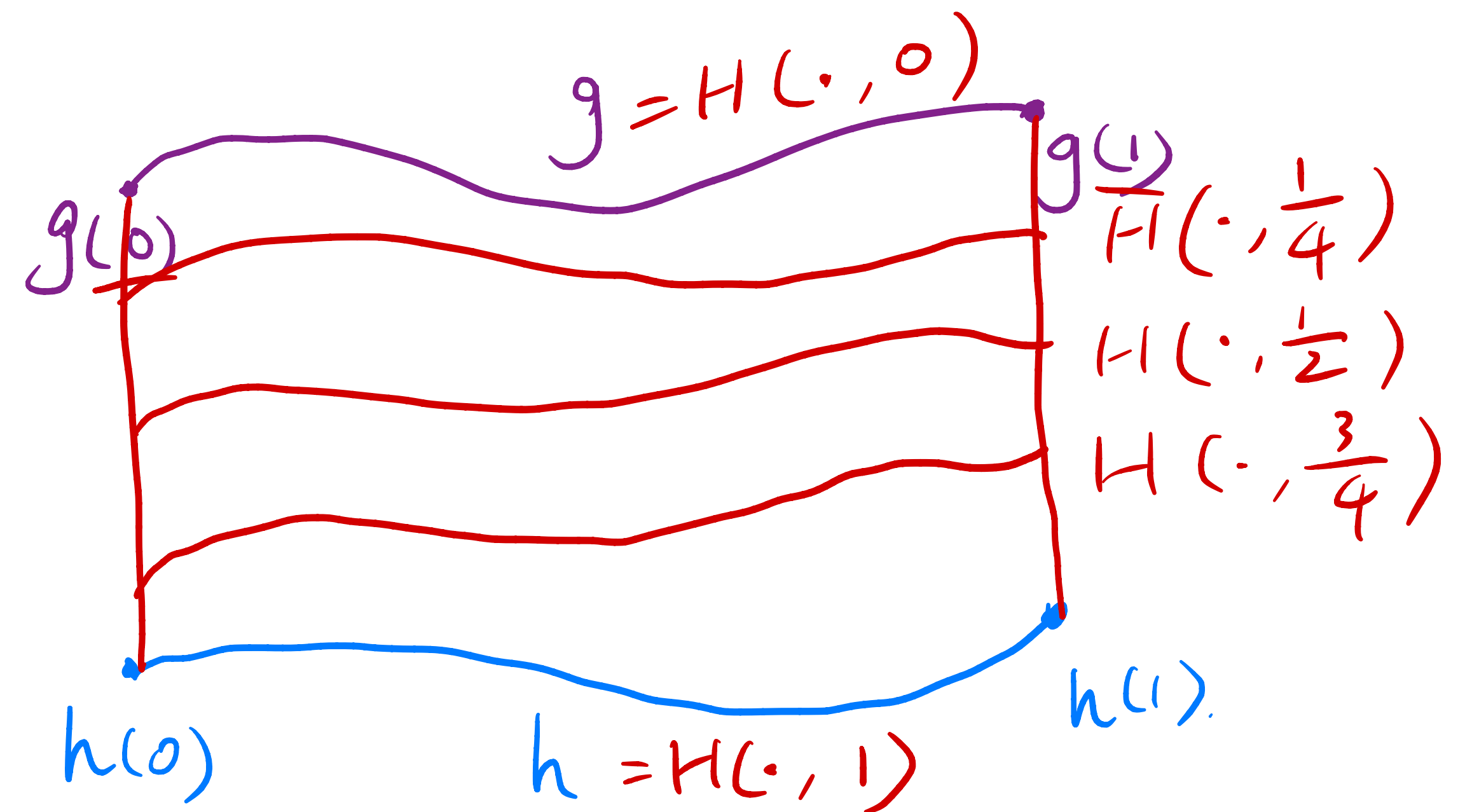
Homotopy

Let $g : X \rightarrow U$ and $h : X \rightarrow U$ be maps.

A homotopy is a map $H : \underline{X} \times \underline{[0,1]} \rightarrow U$ such that $H(\cdot, \underline{0}) = \underline{g}$ and $H(\cdot, \underline{1}) = \underline{f}$ ✓

Two maps are homotopic if there is a homotopy connecting them.

$H : [0,1] \times [0,1] \rightarrow \mathbb{R}^2$
 Can change
 where we are on the curve
 how far along the deformation.



Annulus Example

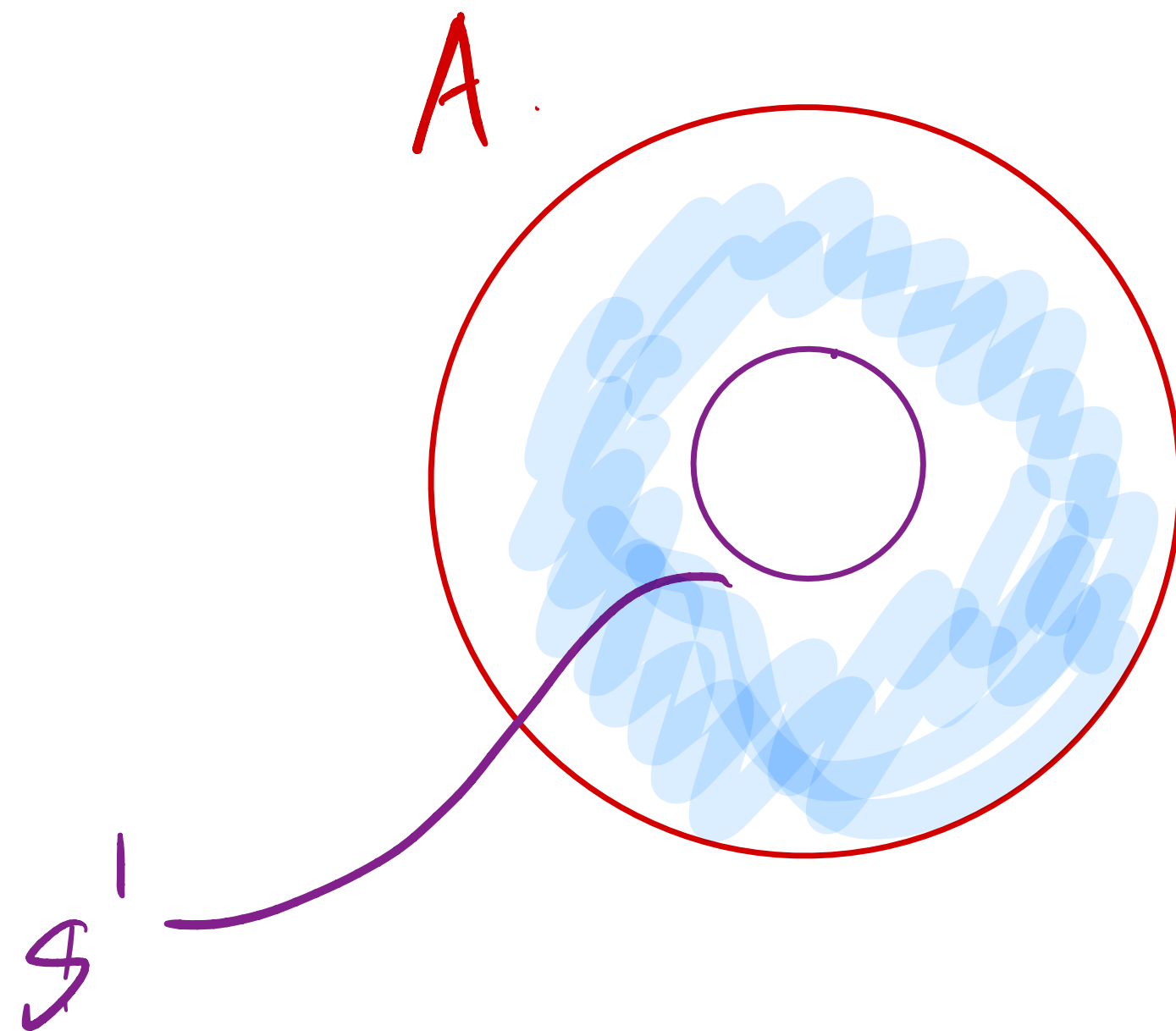
Annulus: $A = \{(\theta, r) \mid 1 \leq r \leq 2\}$. Circle: $\underline{S^1} = \{(\theta, r) \mid r = 1\}$

$g : A \rightarrow \underline{A}$ identity. $h : A \rightarrow A, h(\theta, r) = (\theta, 1)$.

Show $R(\theta, r, t) = (\theta, (1 - t)r + t)$ is a homotopy
 $R : A \times [0, 1] \rightarrow A$.

$$R(\theta, r, 0) = (\theta, r) = g.$$

$$R(\theta, r, 1) = (\theta, 1) = h.$$

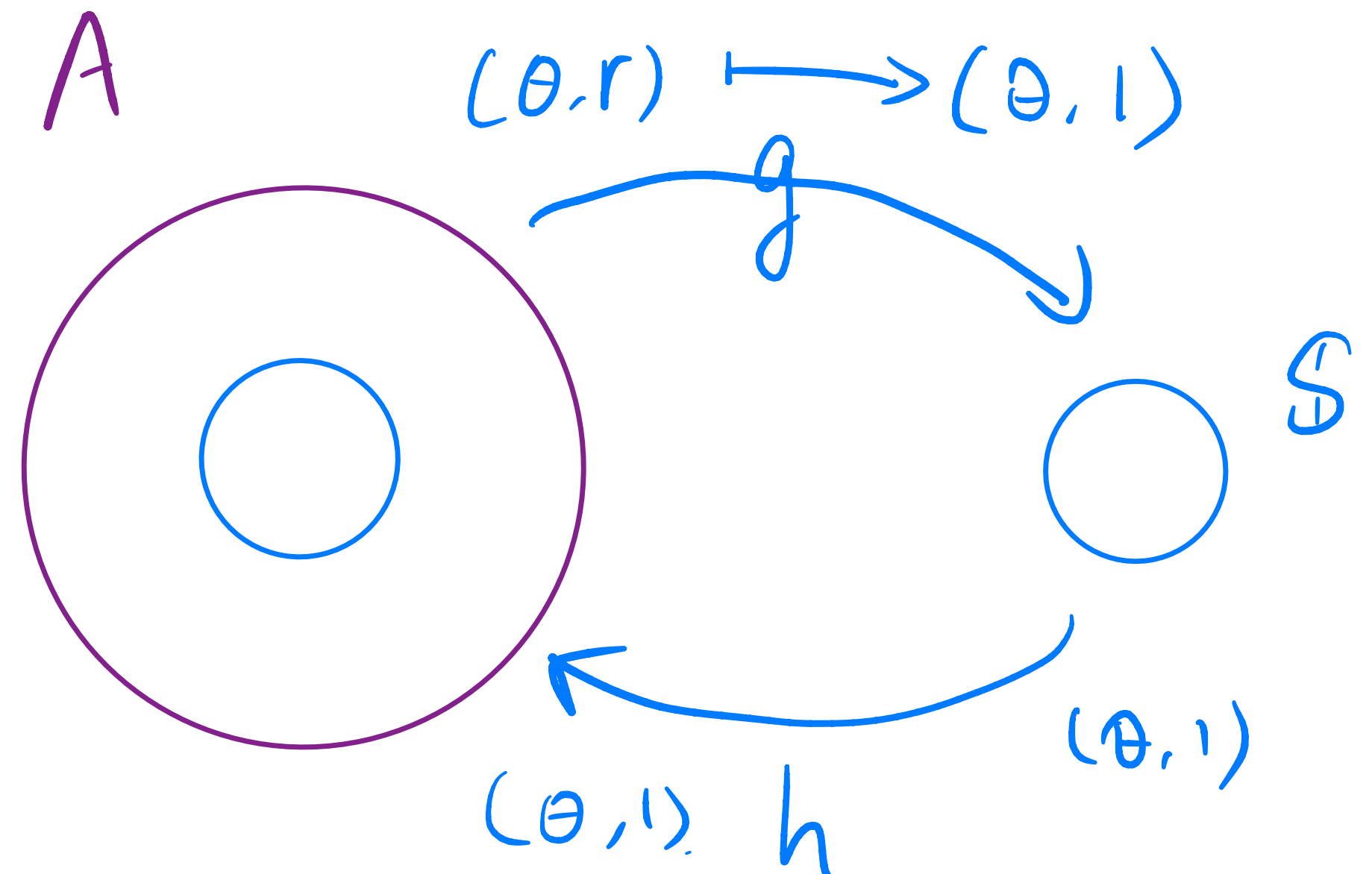


Homotopy Equivalent

Two topological spaces T and U are homotopy equivalent if there exist maps $g : T \rightarrow U$ and $h : U \rightarrow T$ such that $\overbrace{h \circ g} \simeq \text{id}_T$ and $\overbrace{g \circ h} \simeq \text{id}_U$ are homotopic to the appropriate identity maps.

$$\overbrace{h \circ g} \simeq \text{id}_T \quad \overbrace{g \circ h} \simeq \text{id}_U$$

Ex. Show A is homotopy equivalent to $S^1 \subset A$



$$g : A \rightarrow S^1 \quad g(\theta, r) = (\theta, 1)$$

$$h : S^1 \rightarrow A \quad h(\theta, 1) = (\theta, 1)$$

$$1. \quad g \circ h : S^1 \rightarrow S^1$$

$$(\theta, 1) \xrightarrow{h} (\theta, 1) \xrightarrow{g} (\theta, 1)$$

$$\simeq \text{id}_{S^1}$$

$$2. \quad h \circ g : A \rightarrow A$$

$$(\theta, r) \xrightarrow{g} (\theta, 1) \xrightarrow{h} (\theta, 1)$$

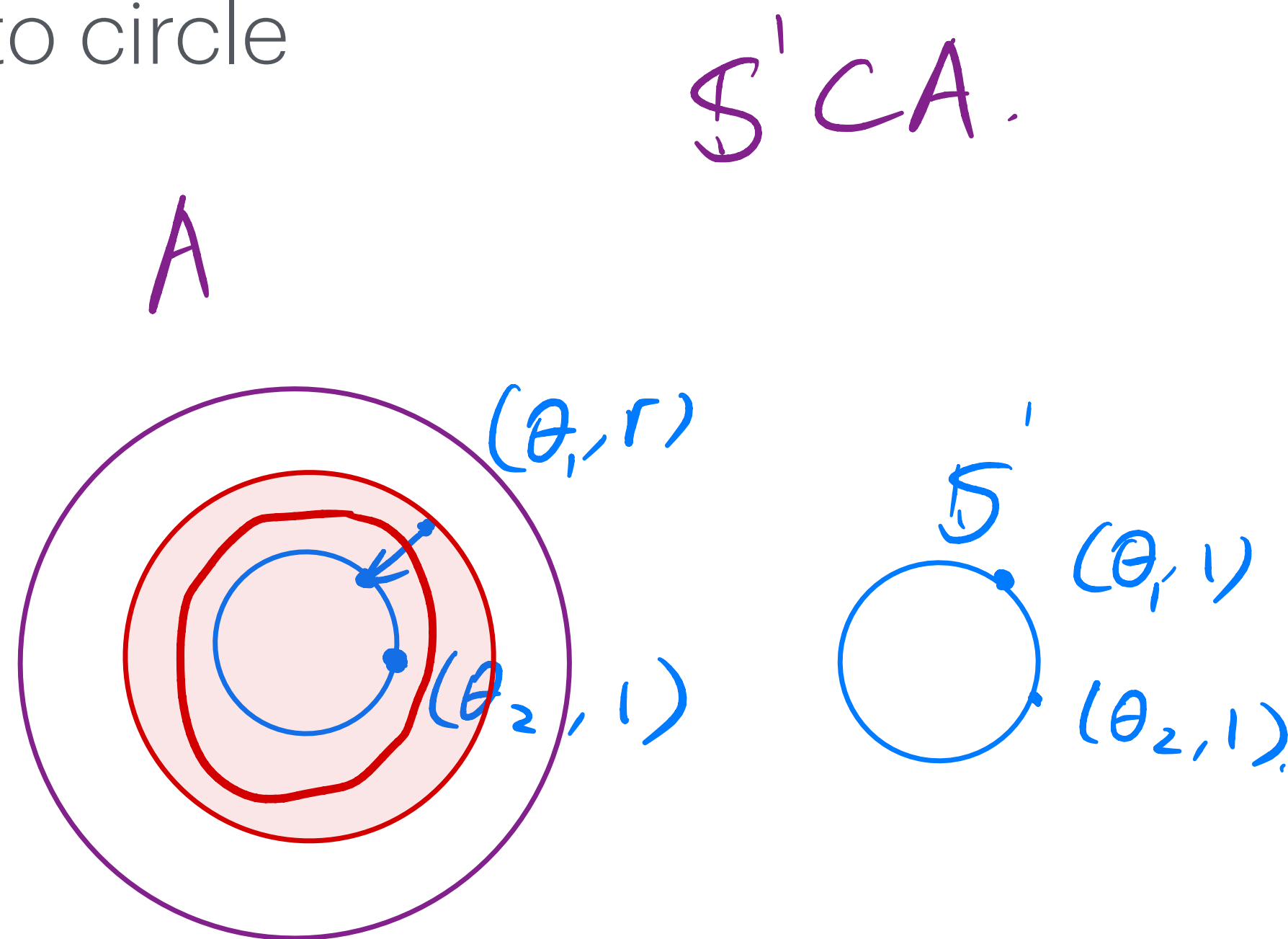
$$\text{id}_A(\theta, r) = (\theta, r)$$

Retract

Let T be a topological space, and let $\underline{U \subset T}$ be a subspace.

A retraction r of T to U is a map $r : T \rightarrow U$ such that $r(x) = x$ for every $x \in U$

Ex. Annulus to circle



$$\underline{r(\theta, 1) = (\theta, 1)}$$

Deformation Retract

The space $U \subseteq T$ is a deformation retract of T if the identity map on T can be continuously deformed to a retraction with no motion of the points already in U .

Specifically, there is a homotopy $R : T \times [0,1] \rightarrow T$ such that

- $R(\cdot, \underline{0})$ is the identity map on T ✓
- $\underline{R(\cdot, 1)}$ is a retraction of T to U ✓
- $R(x, t) = x$ for every $x \in U$ and every $t \in [0,1]$ ✓

Annulus Example

$R(\theta, r, t) = (\theta, (1 - t)r + t)$. Check that R satisfies the three properties to be a deformation retract. $\uparrow$

$$R(\theta, r, 0) = (\theta, r).$$

$$R(\cdot, 0) = \text{id}_A$$

$$R(\theta, r, 1) = \underline{(\theta, 1)}.$$

3. Points on S' never move.

$x \in S'$, $r=1$ then for every $t \in [0, 1]$.

$$R(\theta, 1, t) = (\theta, (1-t) \cdot 1 + t)$$

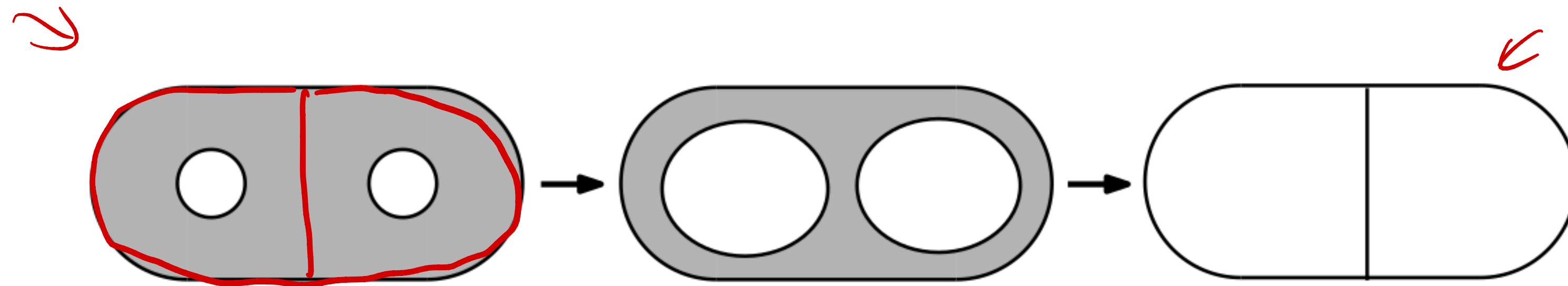
$$= (\theta, 1-t+t)$$

$$= \underline{(\theta, 1)},$$

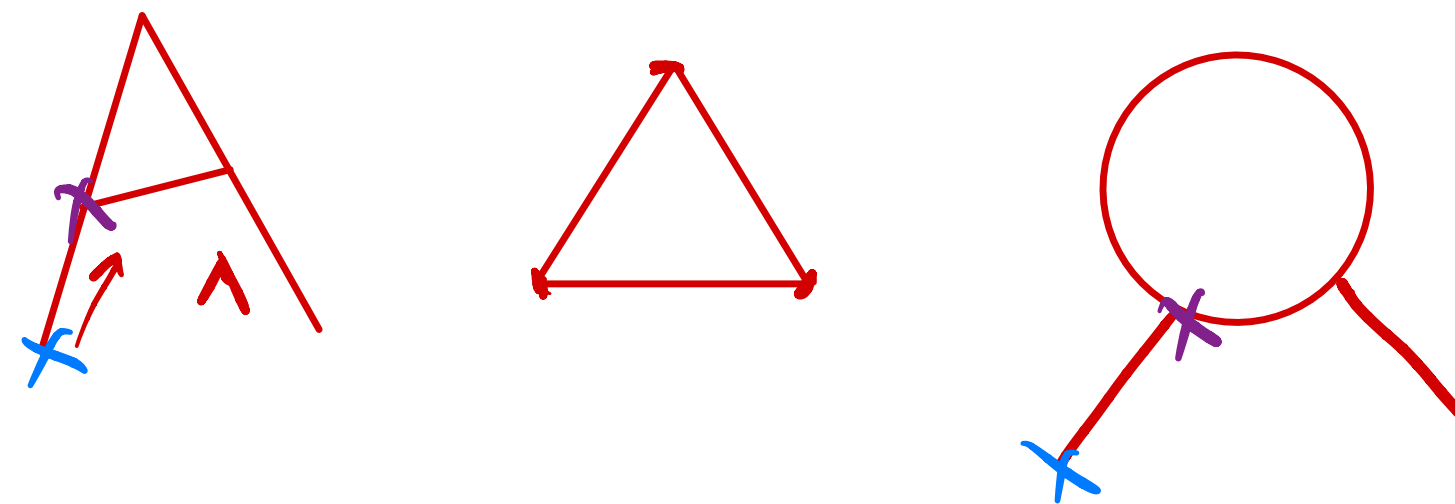
Hence $R(x, t) = x$ for every $x \in S'$

Why do I care?

If U is a deformation retract of T , then T and U are homotopy equivalent.

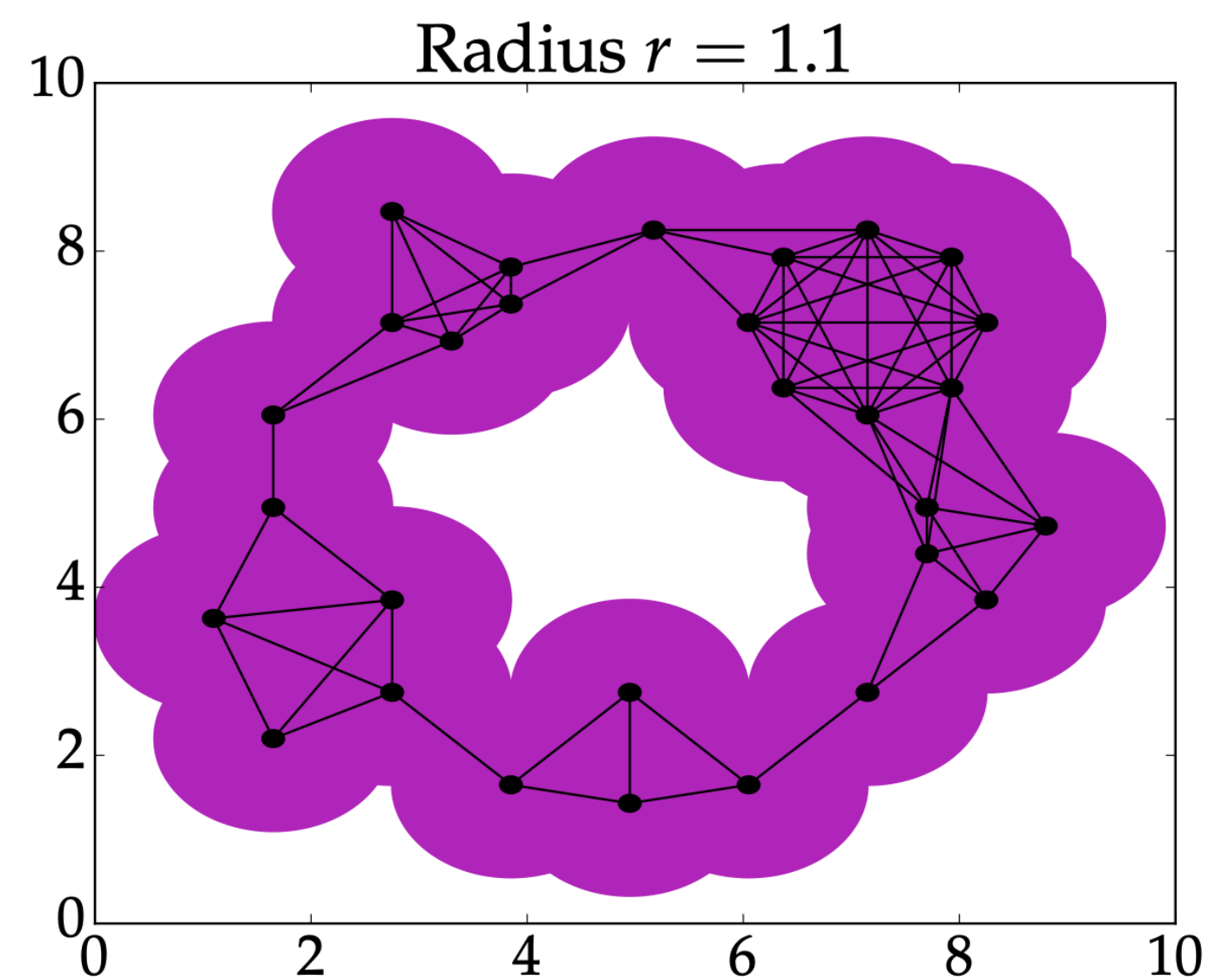
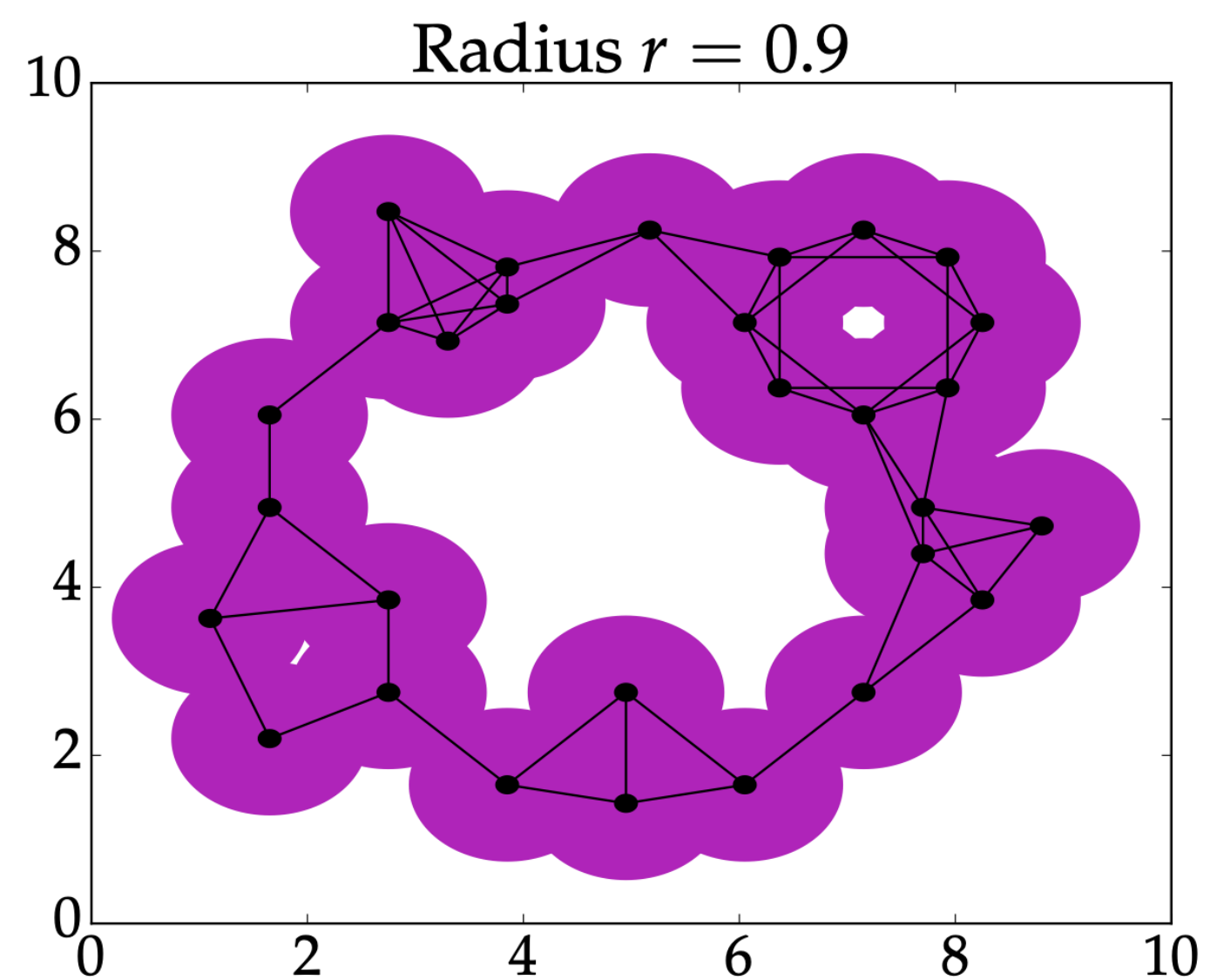
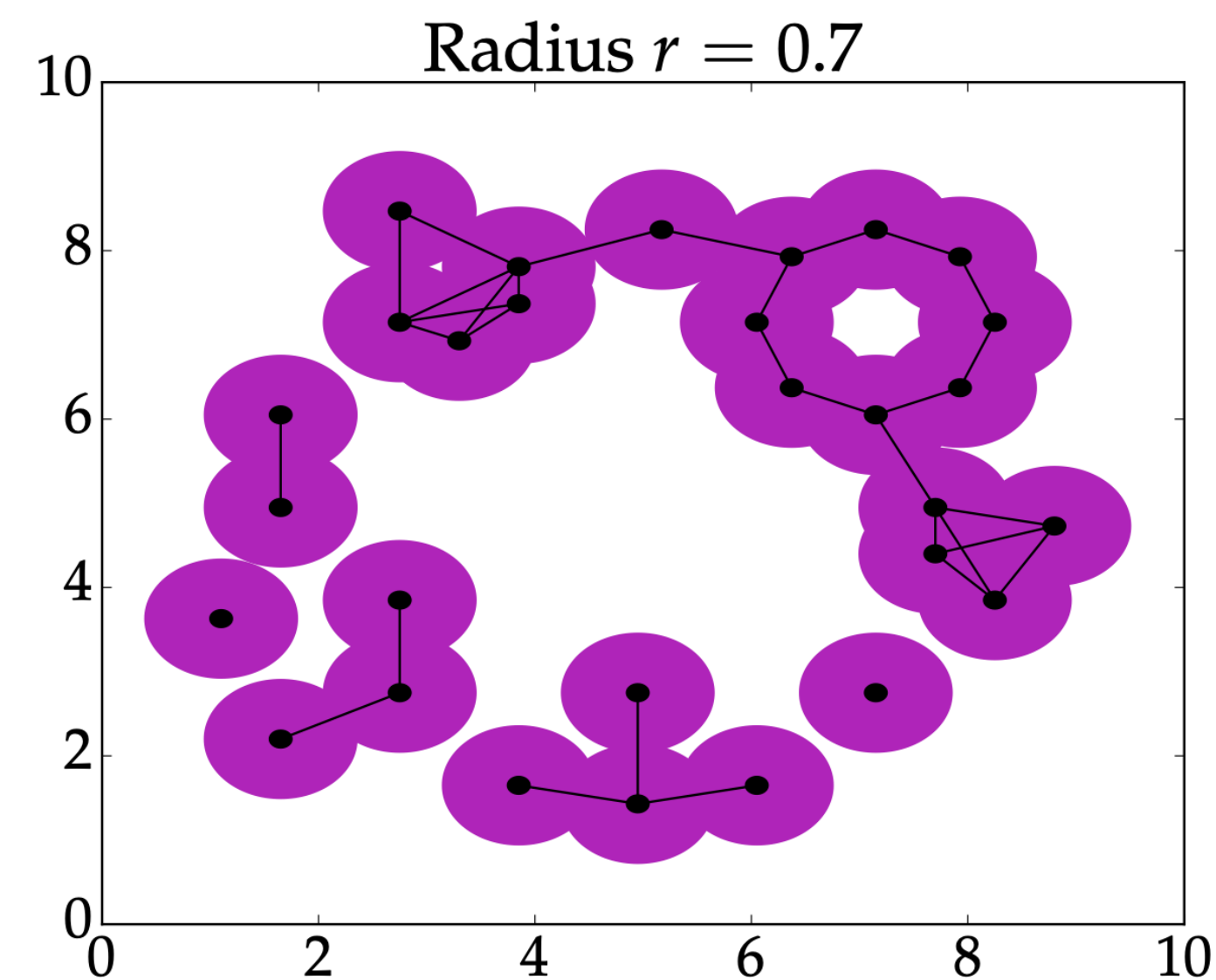
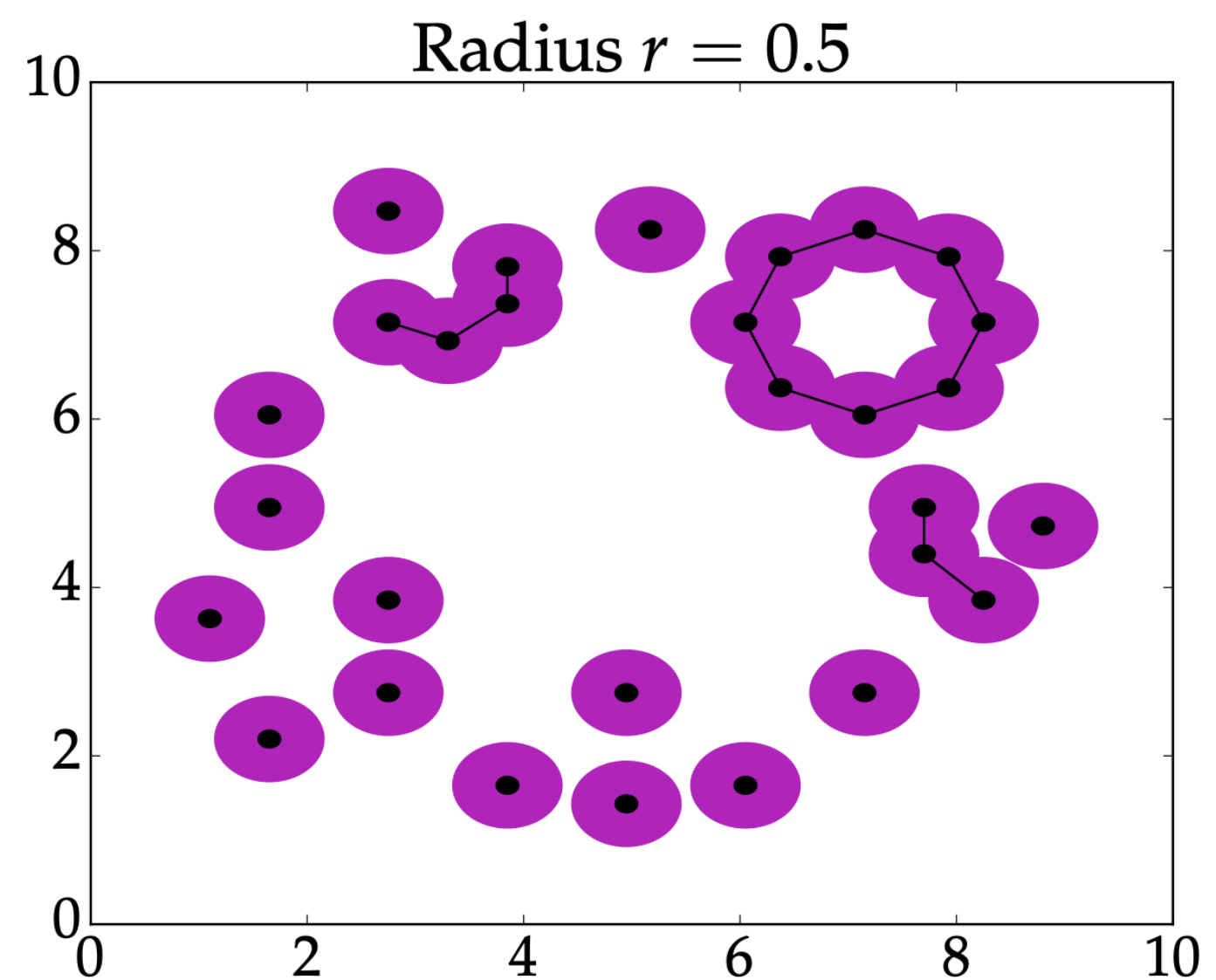
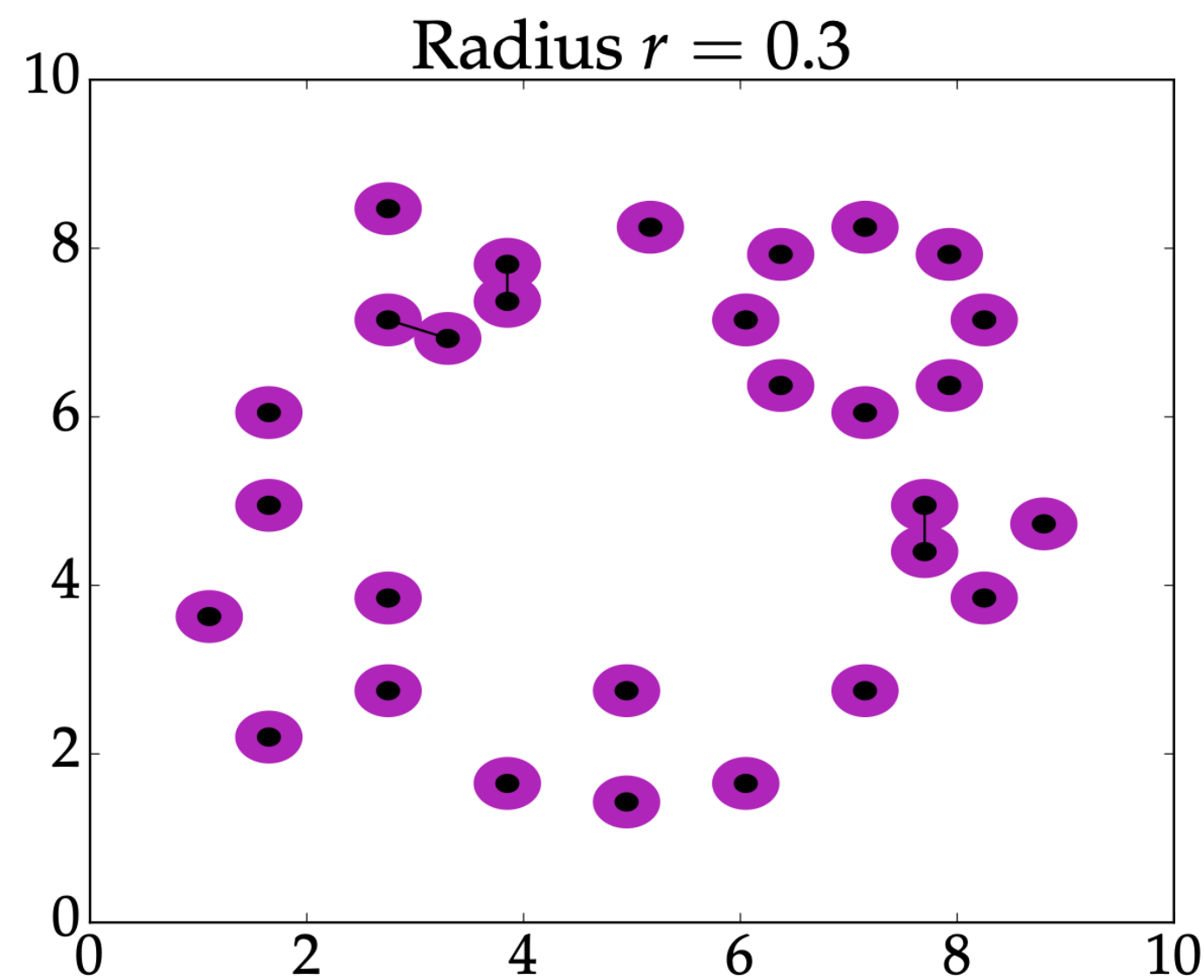


Ex. The letter A to O



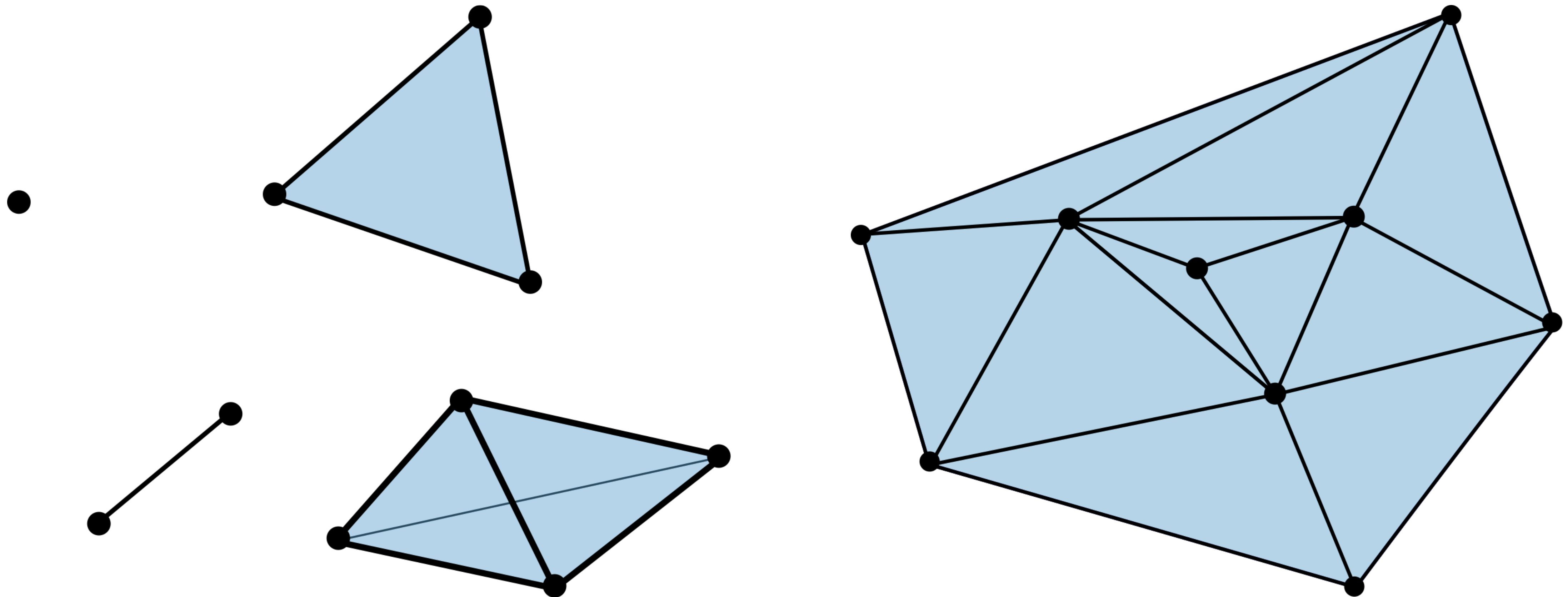
Equivalence classes: subsets of a set containing all elements that are homotopy equivalent.

Goal: what we want to study



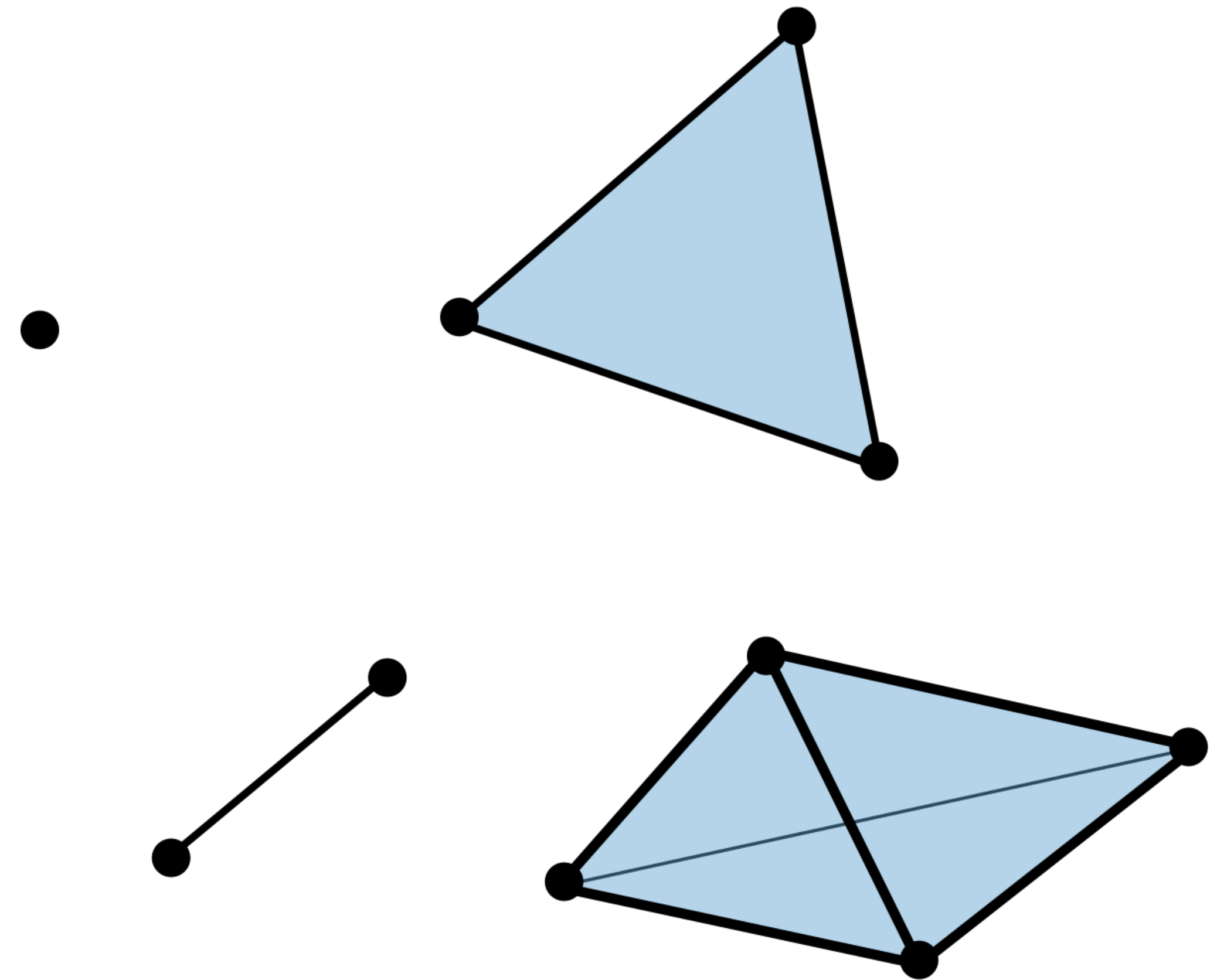
Simplicial Complexes

Idea: turn topology into combinatorics using mathematical “legos”



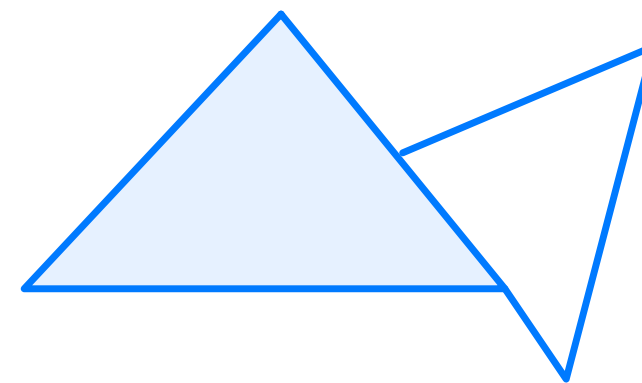
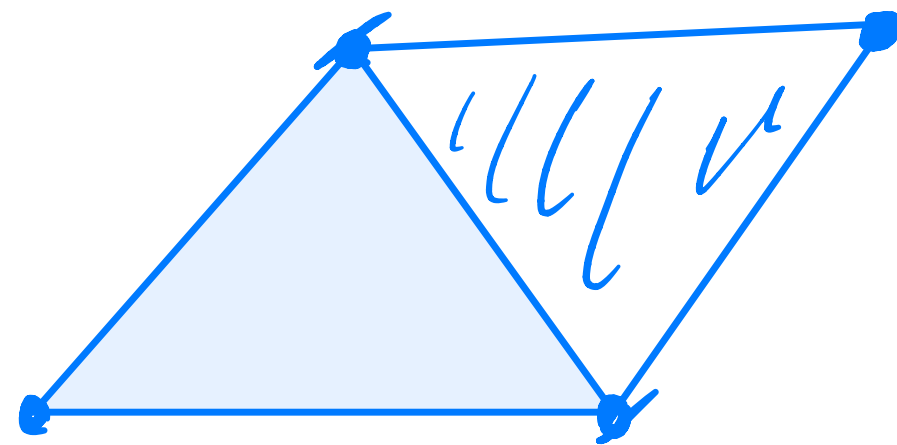
Building Blocks

- 0-simplex: a point (vertex)
- 1-simplex: a line segment (edge) connection two points
- 2-simplex: a triangle (face) filled in between three points
- 3-simplex: a tetrahedron (solid pyramid) filled in between four points
- Higher dimensions: the generalization of a triangle to that dimension



Glueing Rules

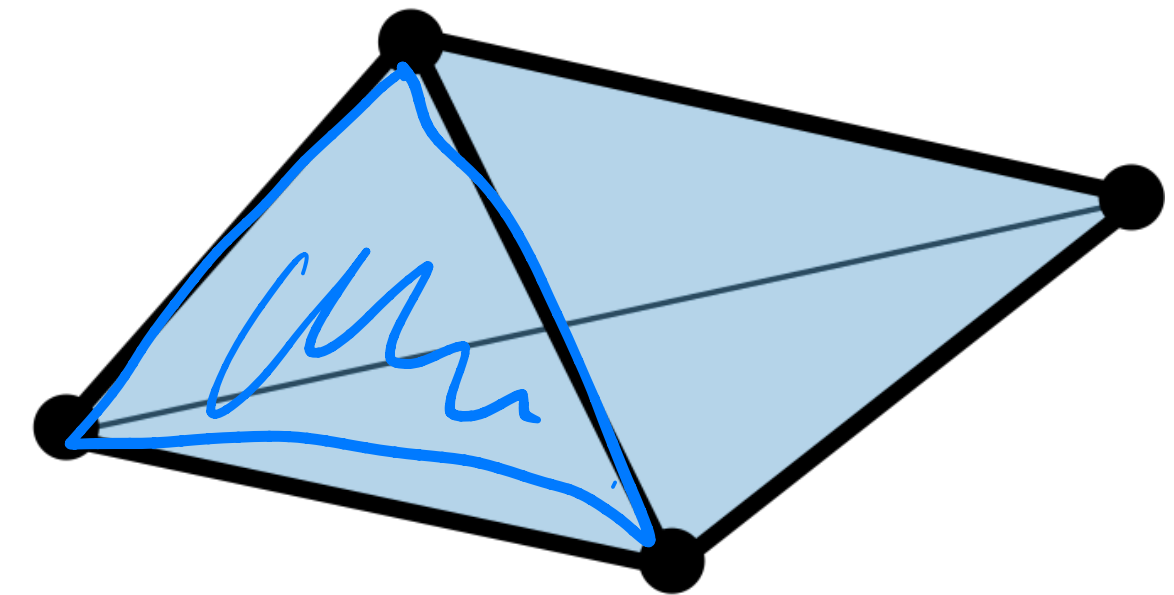
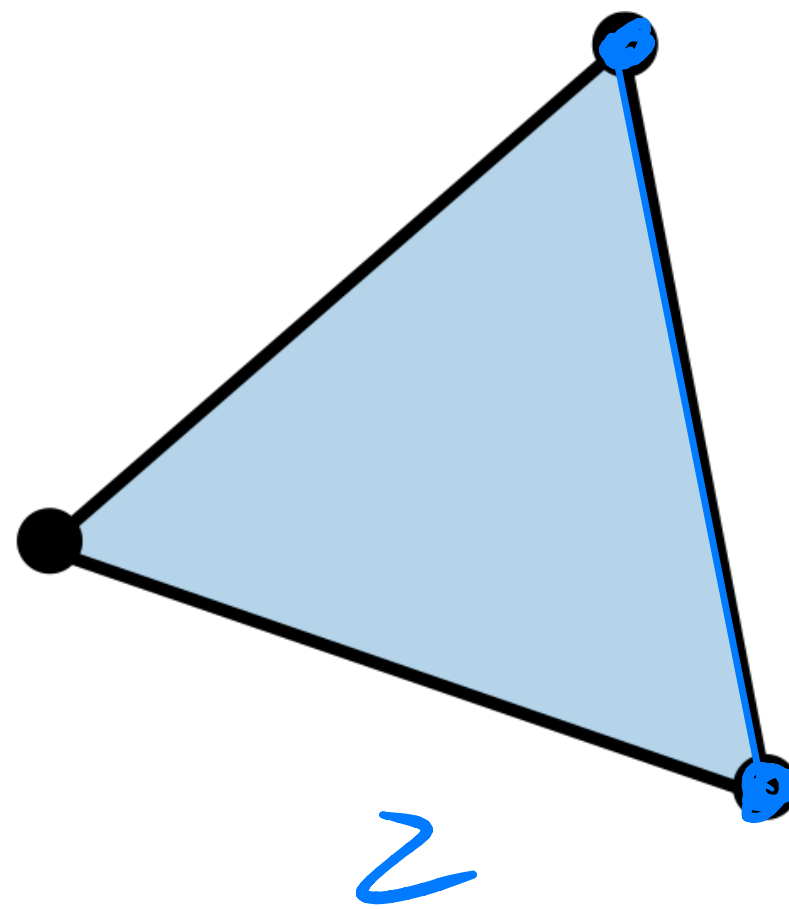
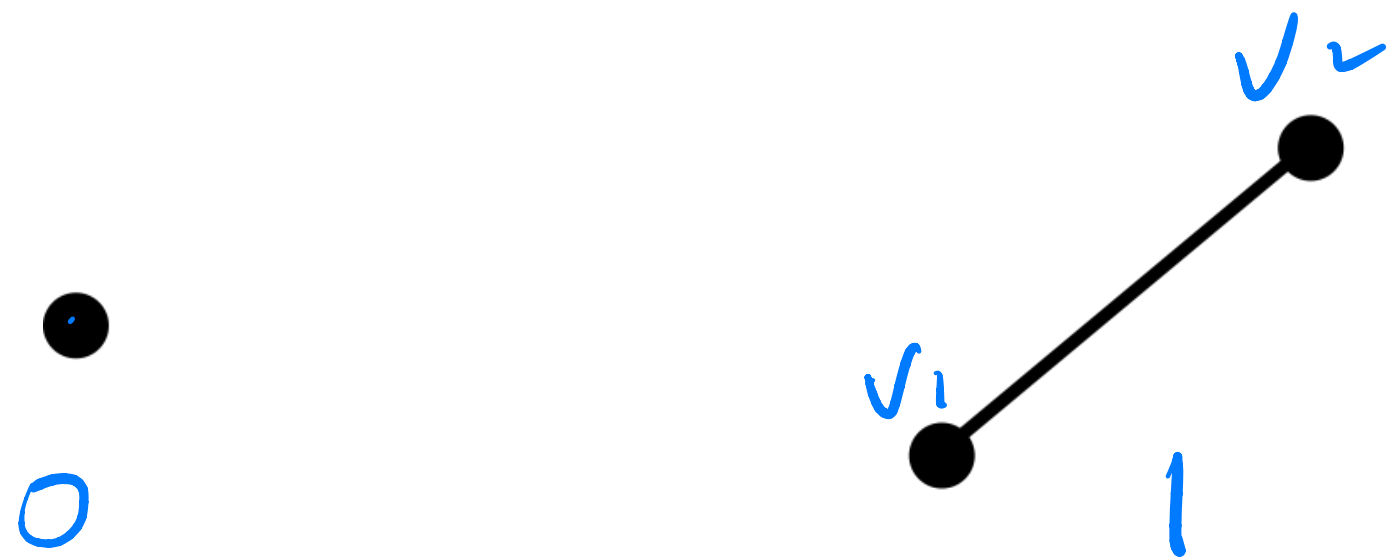
- Face Containment: if a simplex is in the complex, then its subsimplex must be too
 - Given a solid triangle, its three edges and three points must also be part of the complex
- Intersection: If two shapes intersect, their intersection must also be part of the complex



Simplicial Complex: Definition

An n -simplex in $\mathbb{R}^d$ is the convex hull of $n + 1$ affinely independent points in $\mathbb{R}^d$.

nice enough.



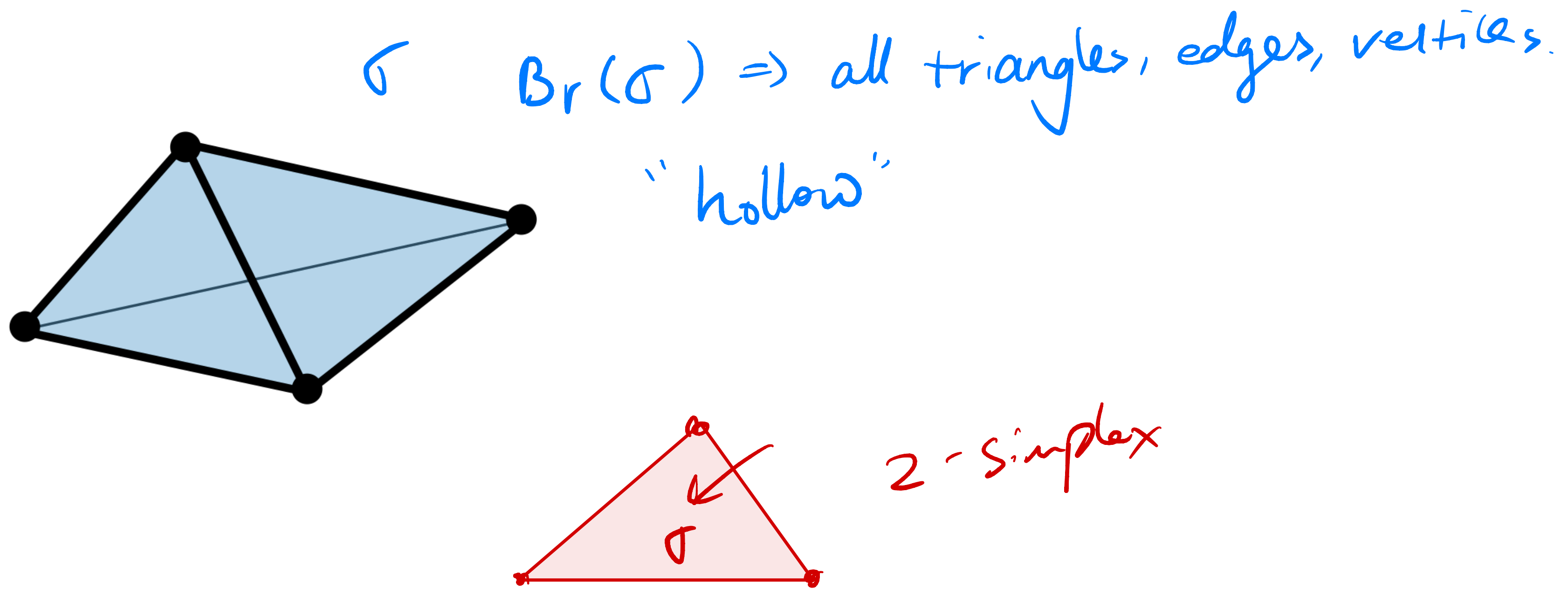
Face: convex hull of subsets of vertices

$\emptyset$ is also face

A face is proper if it isn't the simplex itself

Interior and Boundary

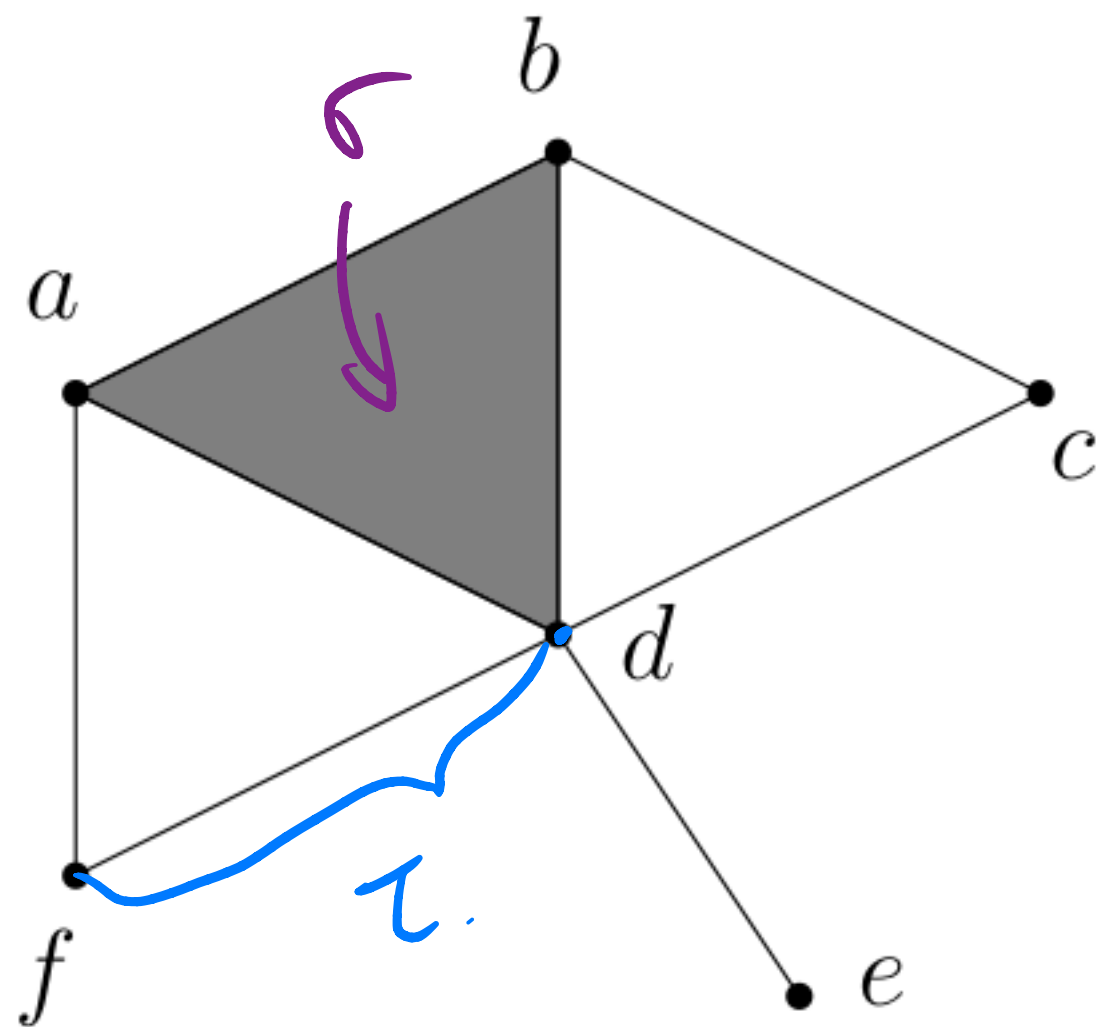
- Let σ be an n -simplex, the union of the proper faces is the boundary of σ , $\text{Bd}(\sigma)$.
- The interior of σ is $\sigma - \text{Bd}(\sigma)$. This is sometimes called the open simplex



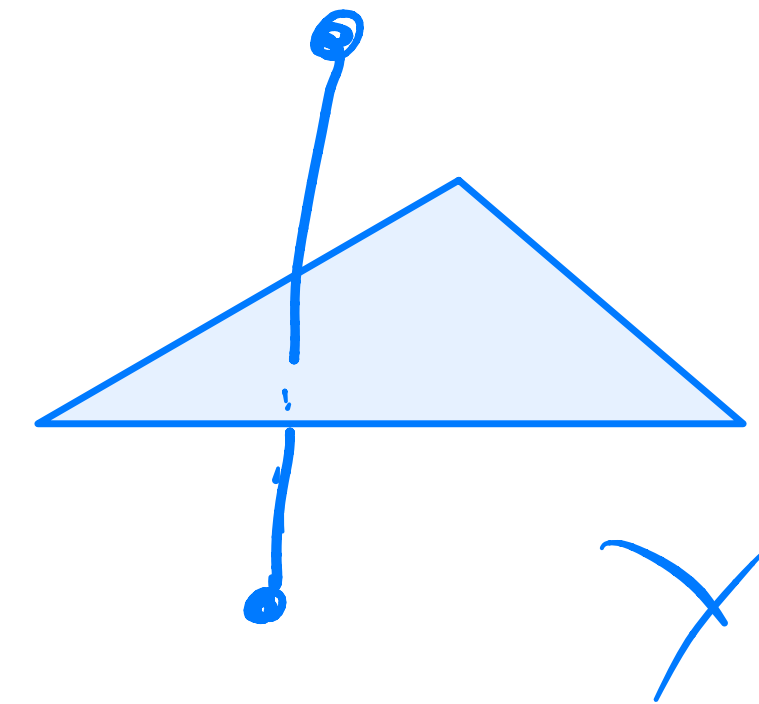
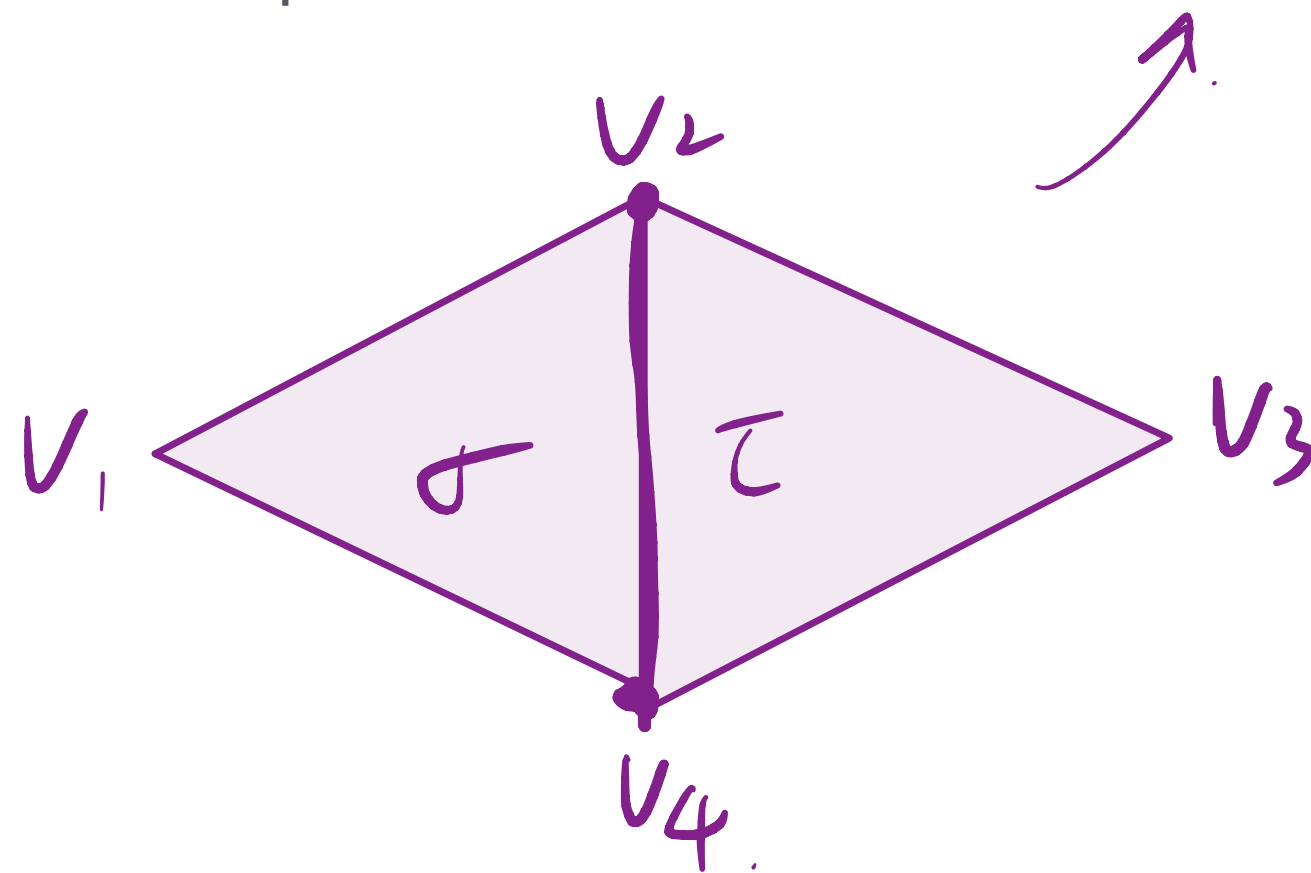
Geometric Simplicial Complex

A geometric simplicial complex K in $\mathbb{R}^d$ is a finite collection of simplices in $\mathbb{R}^d$ such that

- Every face of a simplex of K is in K
- The intersection of any two simplices of K is a face of each.

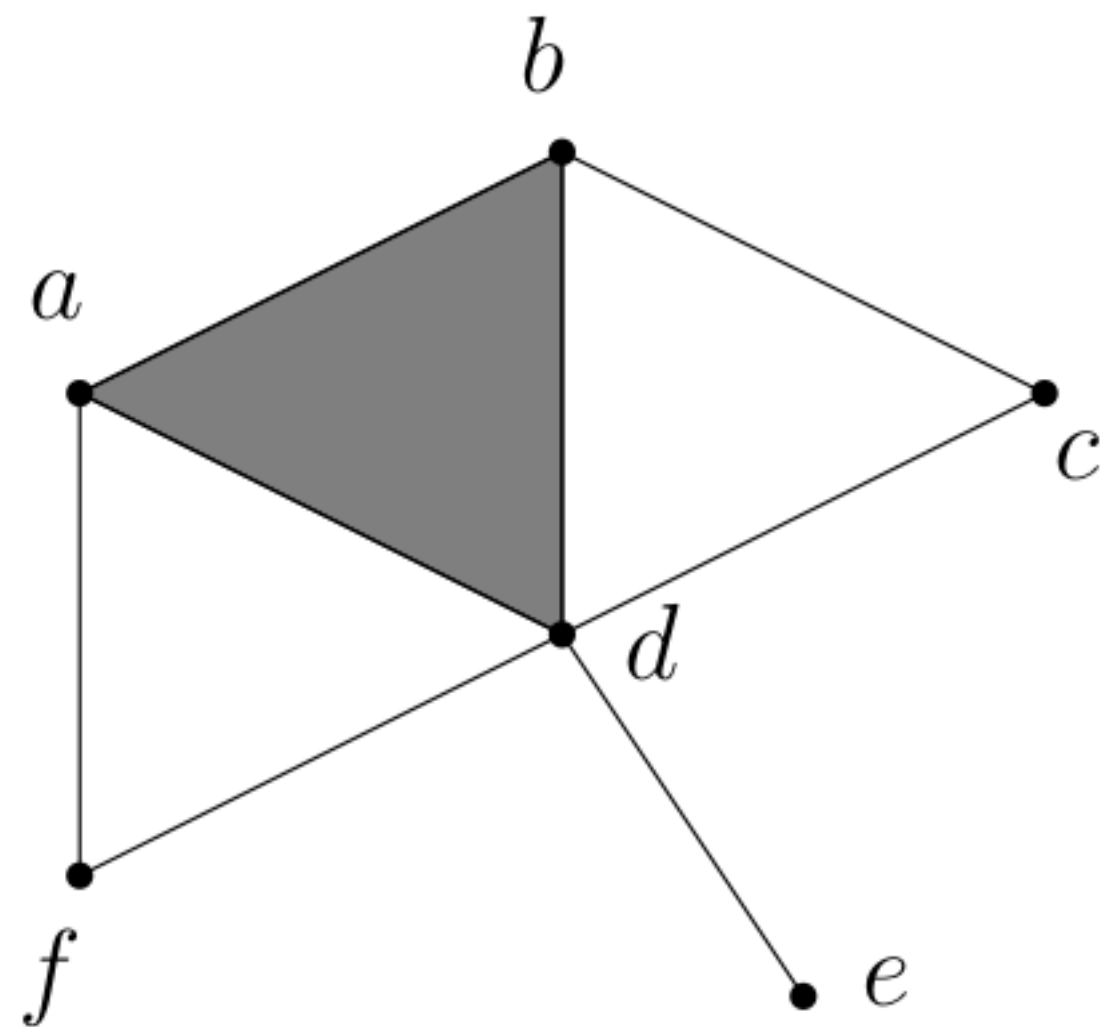


$$\sigma \cap \tau = d.$$



Abstract Simplicial Complex

An abstract simplicial complex $\mathcal{K}$ is a finite collection of finite non-empty sets of V such that $\alpha \in \mathcal{K}$ and $\underline{\beta} \subseteq \underline{\alpha}$ implies $\beta \in \mathcal{K}$. $\leftarrow$



$$V = \{a, b, c, d, e, f\}$$

$$\mathcal{K} = \{\{a\}, \{b\}, \dots, \{f\}, \{a, b\}, \{b, c\}, \{c, d\}, \{b, d\}, \{a, d\}, \{f, d\}, \{a, f\}, \{d, e\}, \underbrace{\{a, b, d\}}_{\alpha}, \underbrace{\{a, d\}}_{\beta}\}$$

$\beta \subseteq \alpha$ ✓

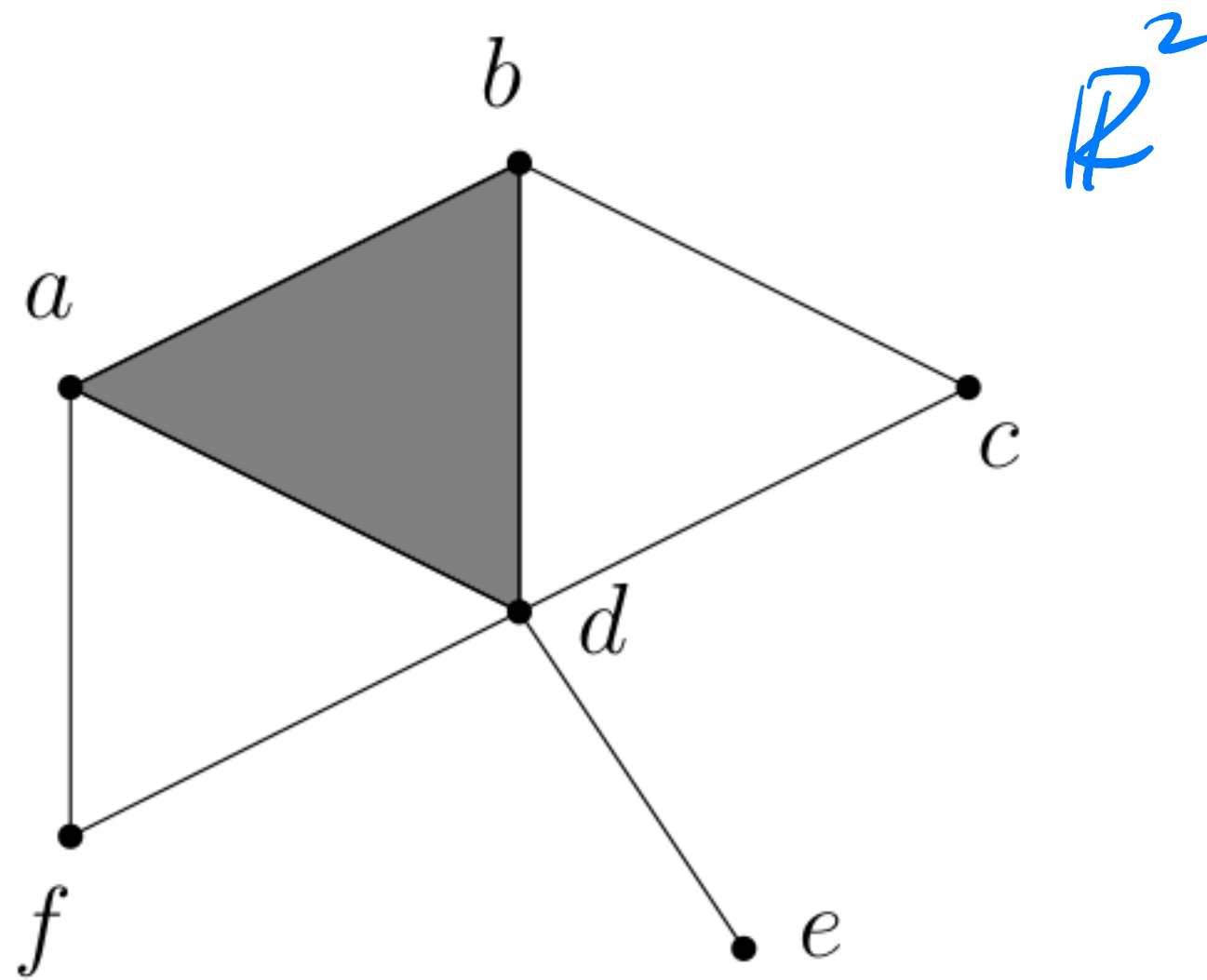
$$\mathcal{K} = \{ \{a\}, \{b\}, \{c\}, \underbrace{\{a, b, c\}}_{\alpha} \} \leftarrow$$

$\times$

$$\{a, b\} = \beta, \beta \subseteq \alpha, \beta \notin \mathcal{K}$$

Geometric Realizations

Geometric simplicial complex K is geometric realization of abstract simplicial complex $\mathcal{K}$ if there is an embedding $f: V(\mathcal{K}) \rightarrow \mathbb{R}^d$ that takes every abstract n -simplex $\{v_0, \dots, v_n\}$ in $\mathcal{K}$ to the geometric k -simplex that is the convex hull of $f(v_0), \dots, f(v_n)$.



Theorem: Every n -dimensional simplicial complex has a non-trivial geometric realization in $\mathbb{R}^{2n+1}$ ✓

Underlying Space

$|K|$ is the underlying space of K , $|K| = \bigcup_{\sigma \in K} \underline{\sigma}$

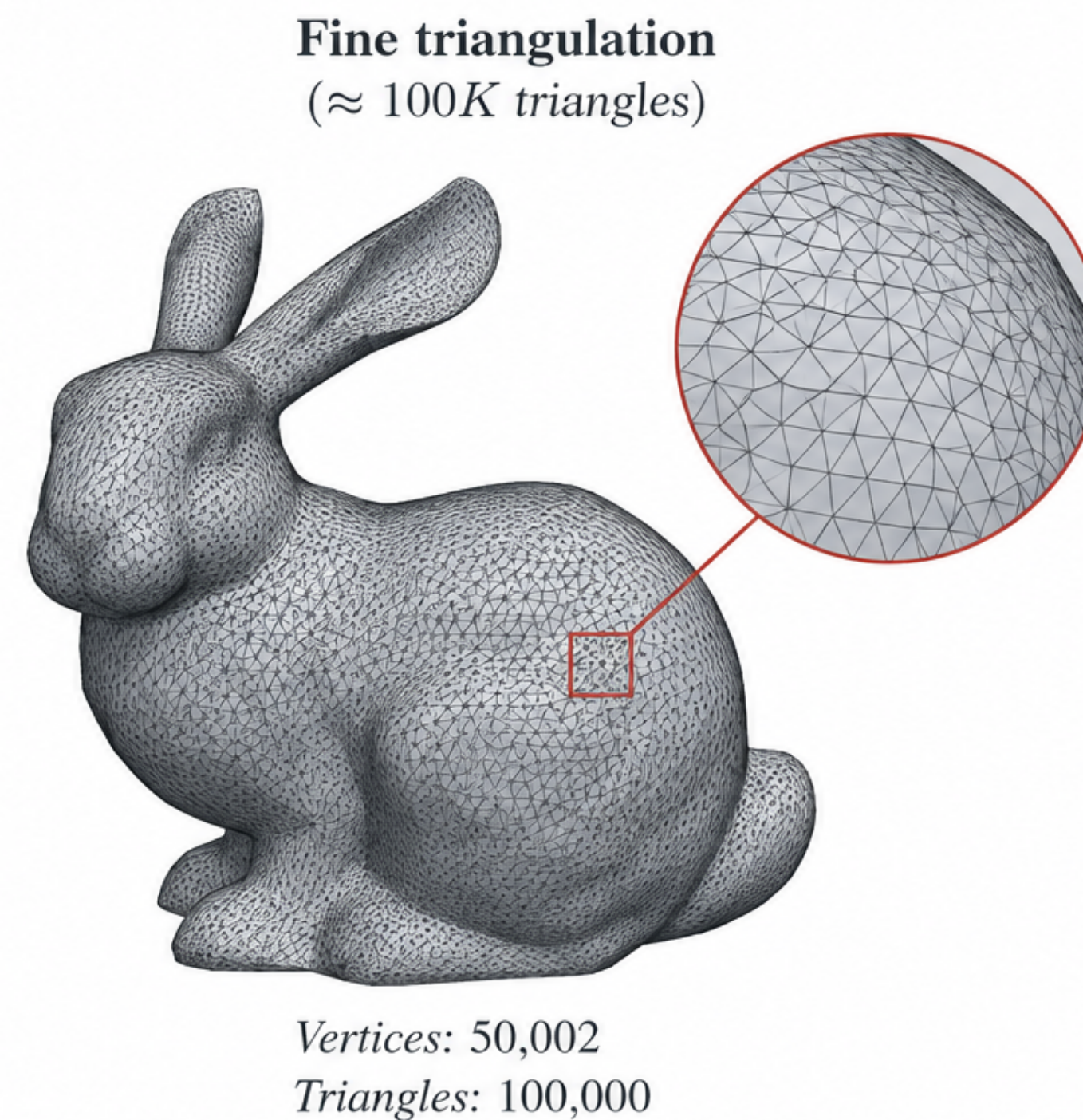
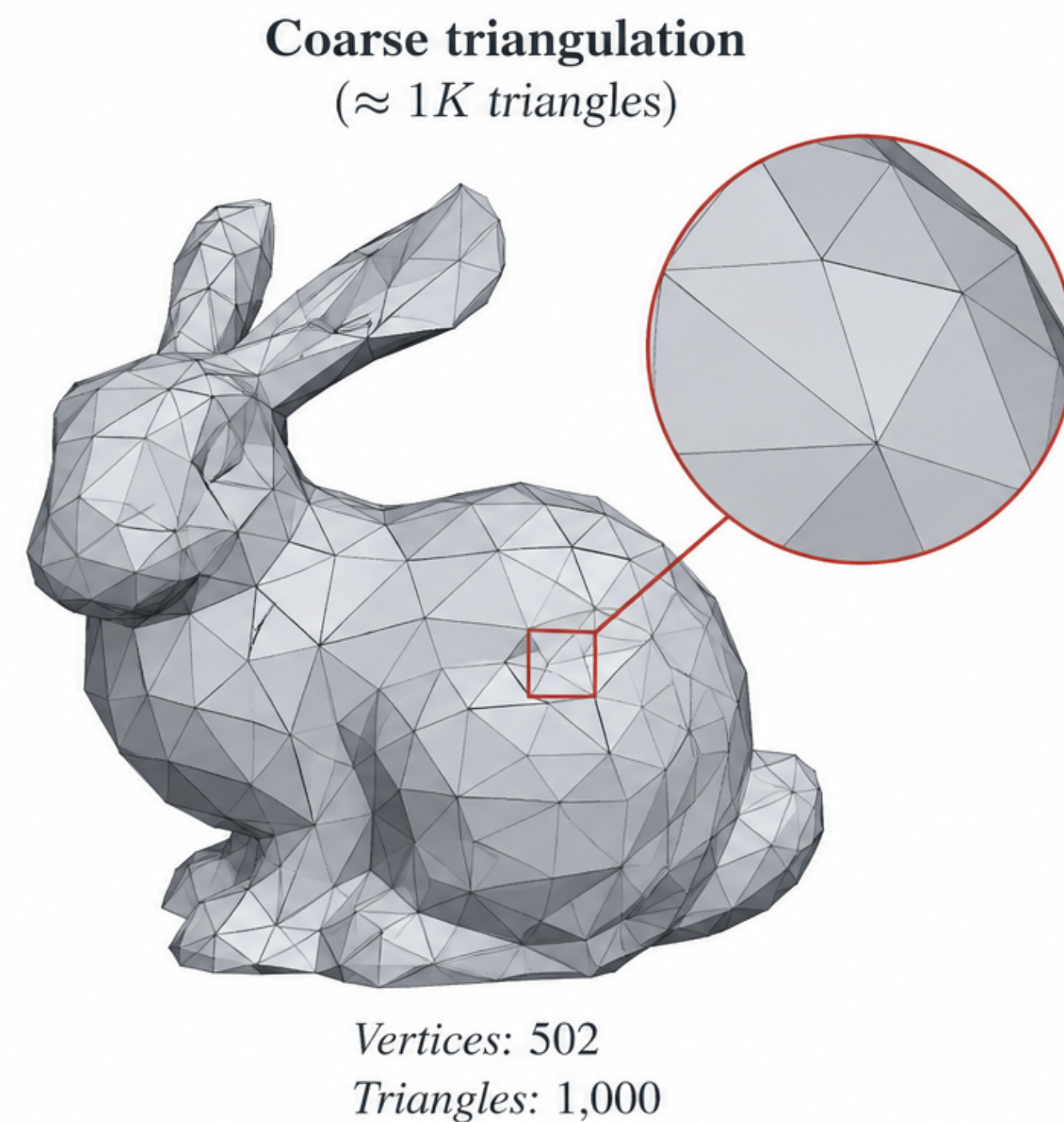
Ex. K = $\{\{v_0\}, \{v_1\}, \{v_2\}, \{v_0, v_1\}, \{v_1, v_2\}, \{v_0, v_2\}, \{v_0, v_1, v_2\}\}$

Triangulations

A simplicial complex K is a triangulation of a topological space T if $|K|$ is homeomorphic to T .

Note: Not all spaces are triangulable;

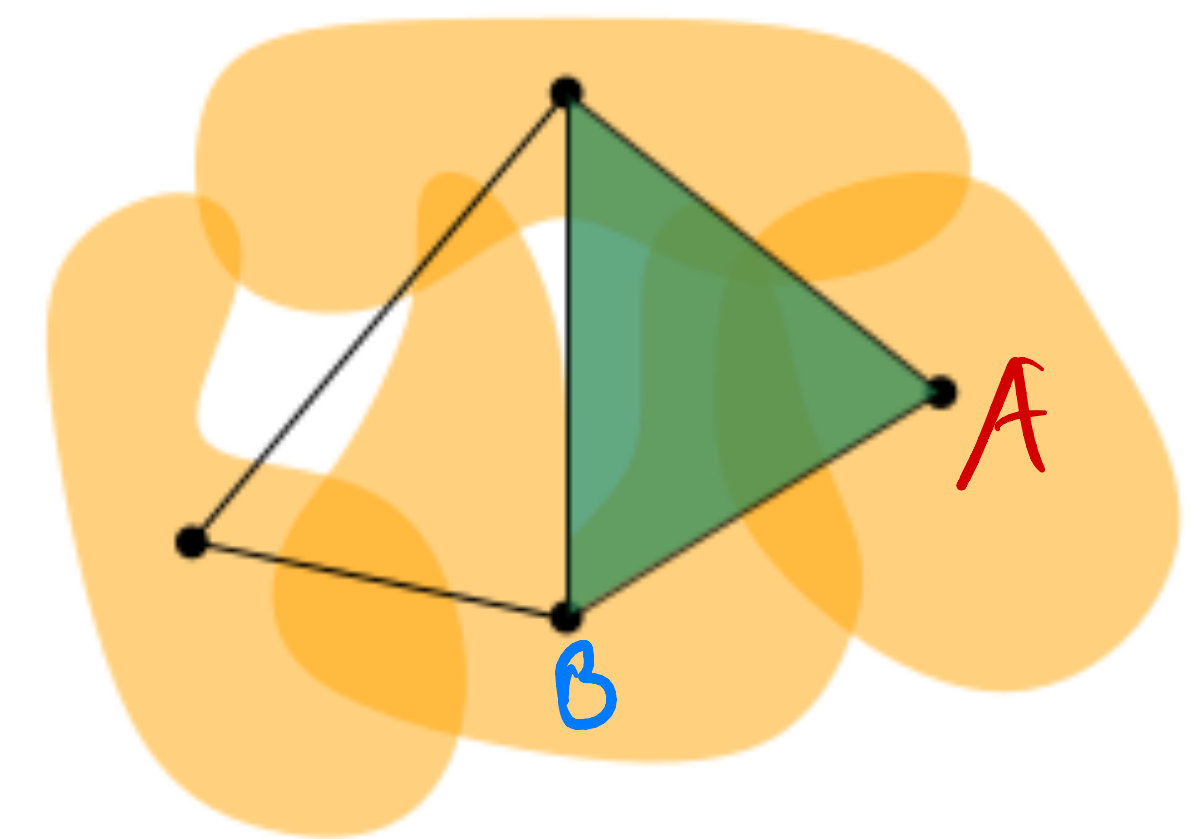
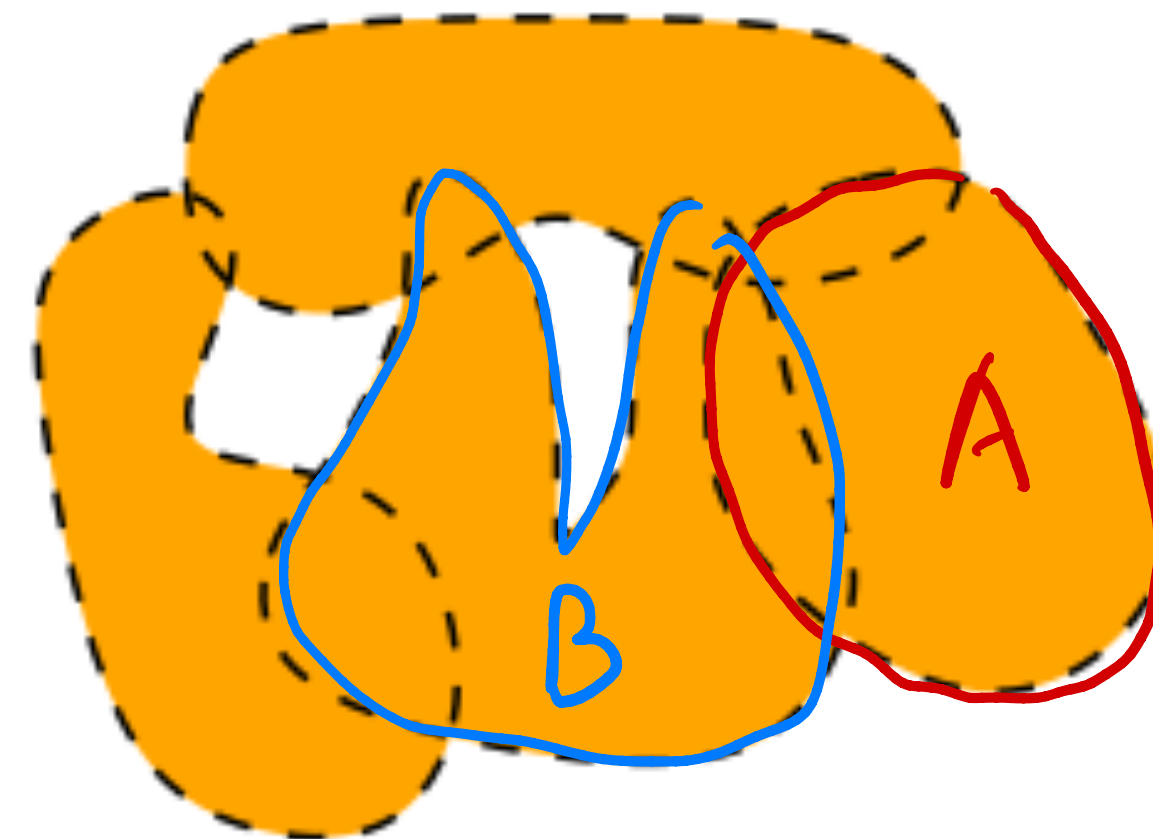
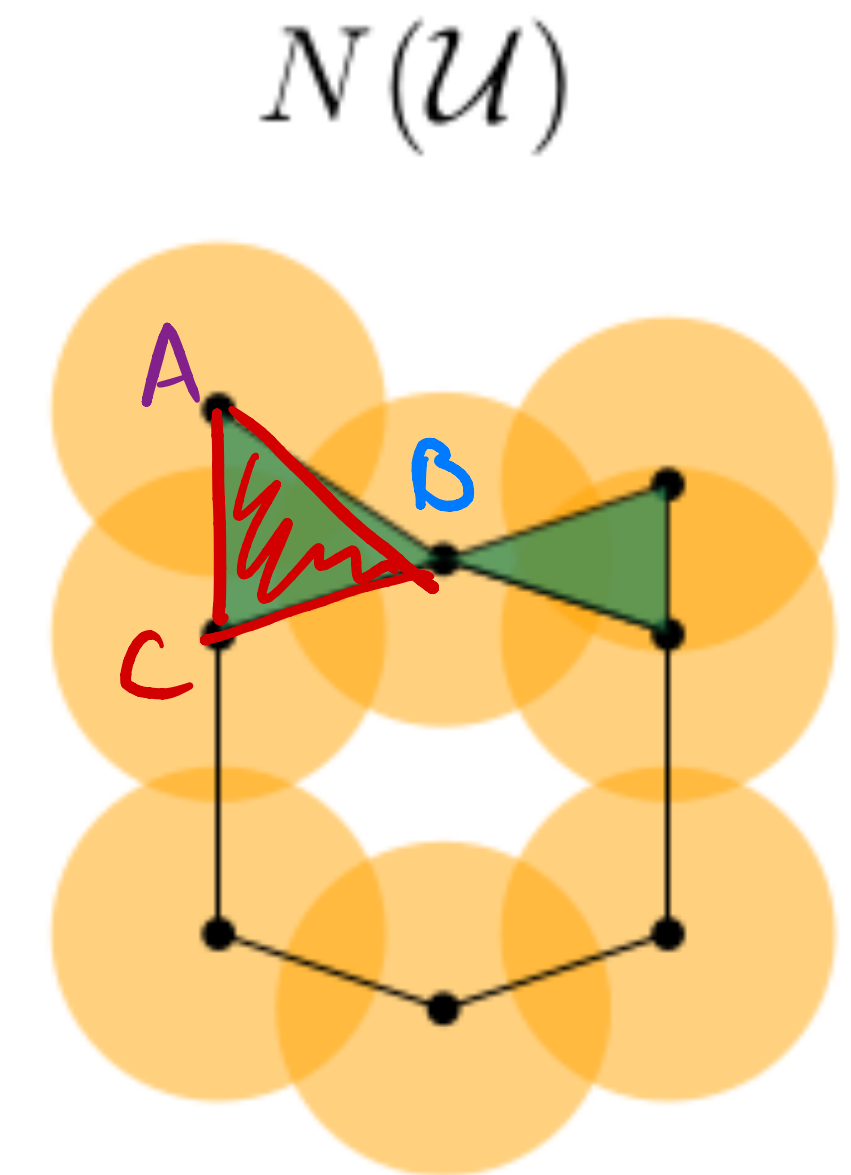
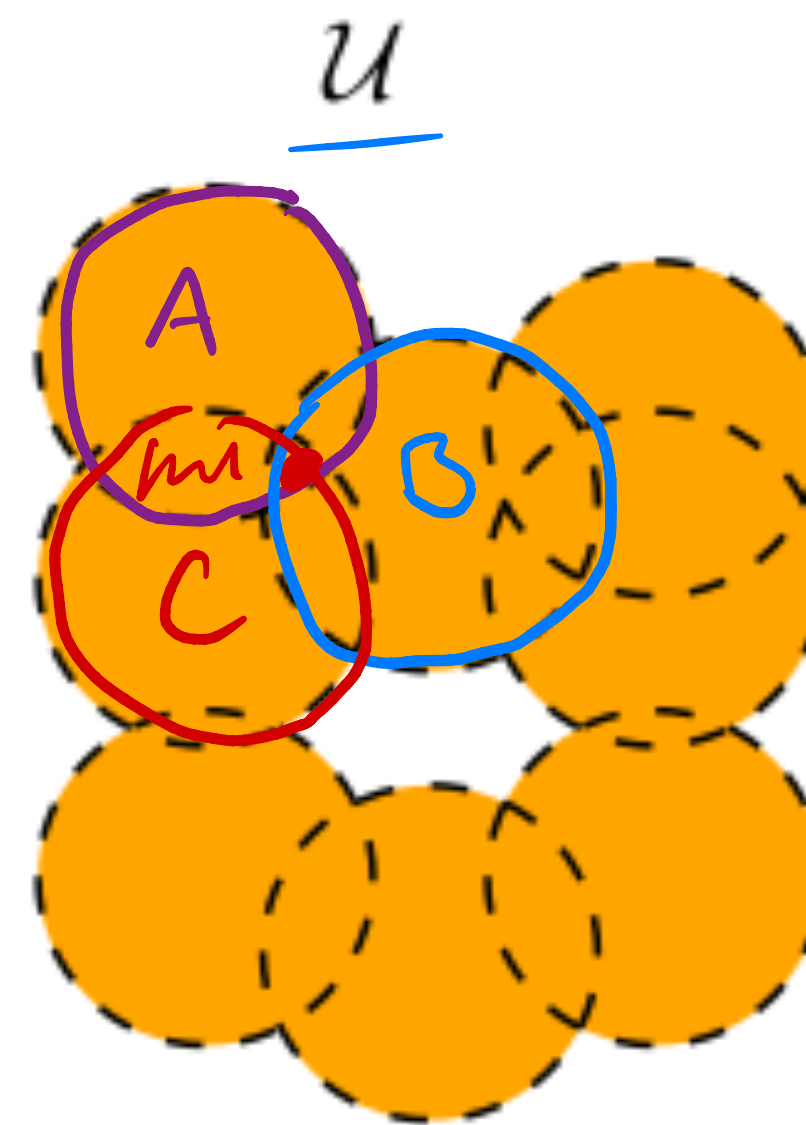
A triangulable space has infinitely many triangulations.



Nerve

Given a finite collection of sets $\mathcal{U}$, the nerve is the simplicial complex

$$\text{Nrv}(\mathcal{U}) = \{X \subseteq \mathcal{U} \mid \bigcap_{U \in X} U \neq \emptyset\}$$



Check: This is actually an abstract simplicial complex!

Recall: K is a simplicial complex if $\sigma \in K$ and $\tau \leq \sigma$ implied $\tau \in K$

$$\text{Nrv}(\mathcal{U}) = \{X \subseteq \mathcal{U} \mid \bigcap_{U \in X} U \neq \emptyset\}$$

Suppose $\underline{X} \in \text{Nrv}(\mathcal{U}) \Rightarrow \bigcap_{U \in X} U \neq \emptyset$.

take any face $\underline{Y} \subseteq \underline{X}$, need to show $Y \in \text{Nrv}(\mathcal{U})$

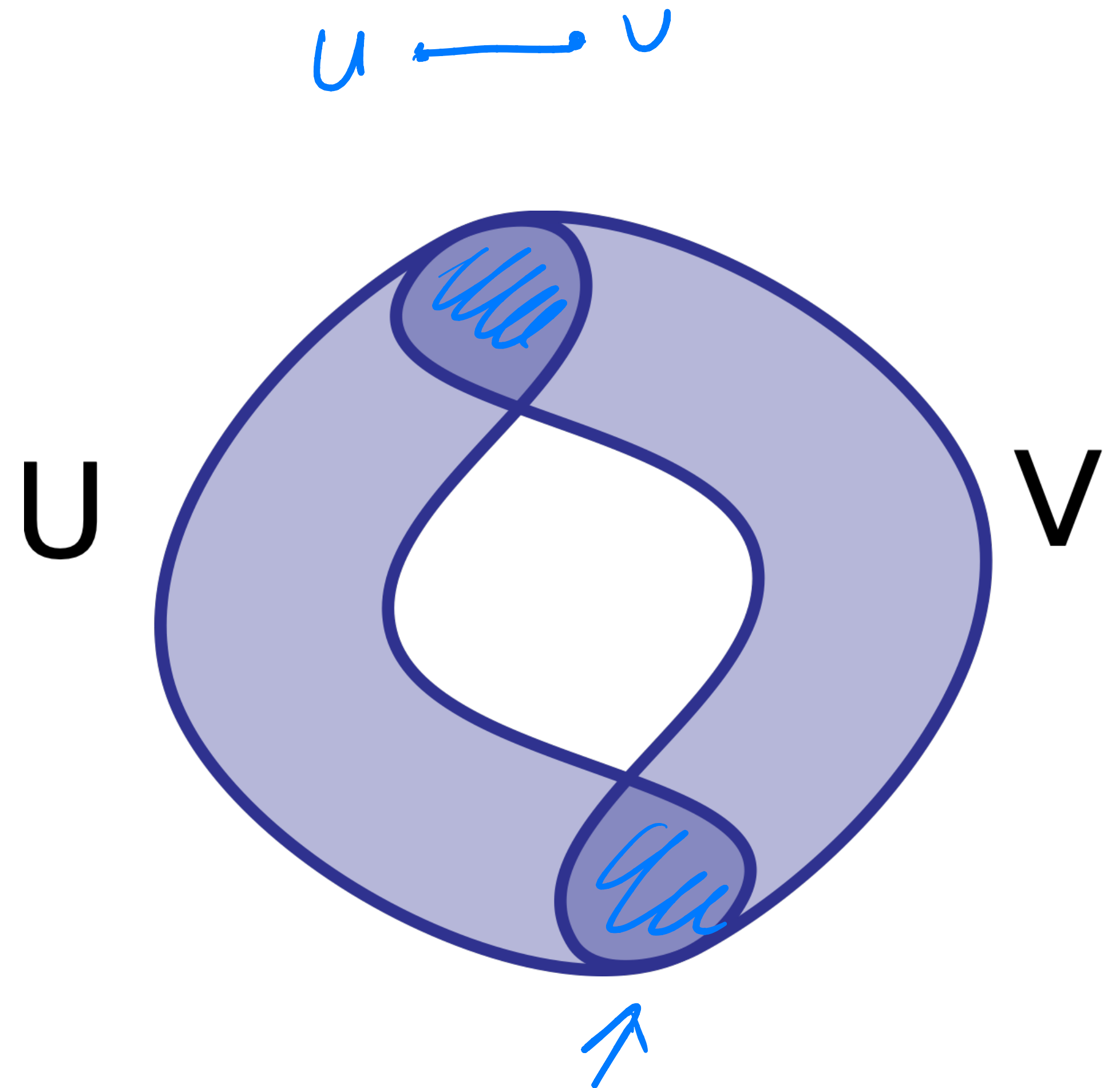
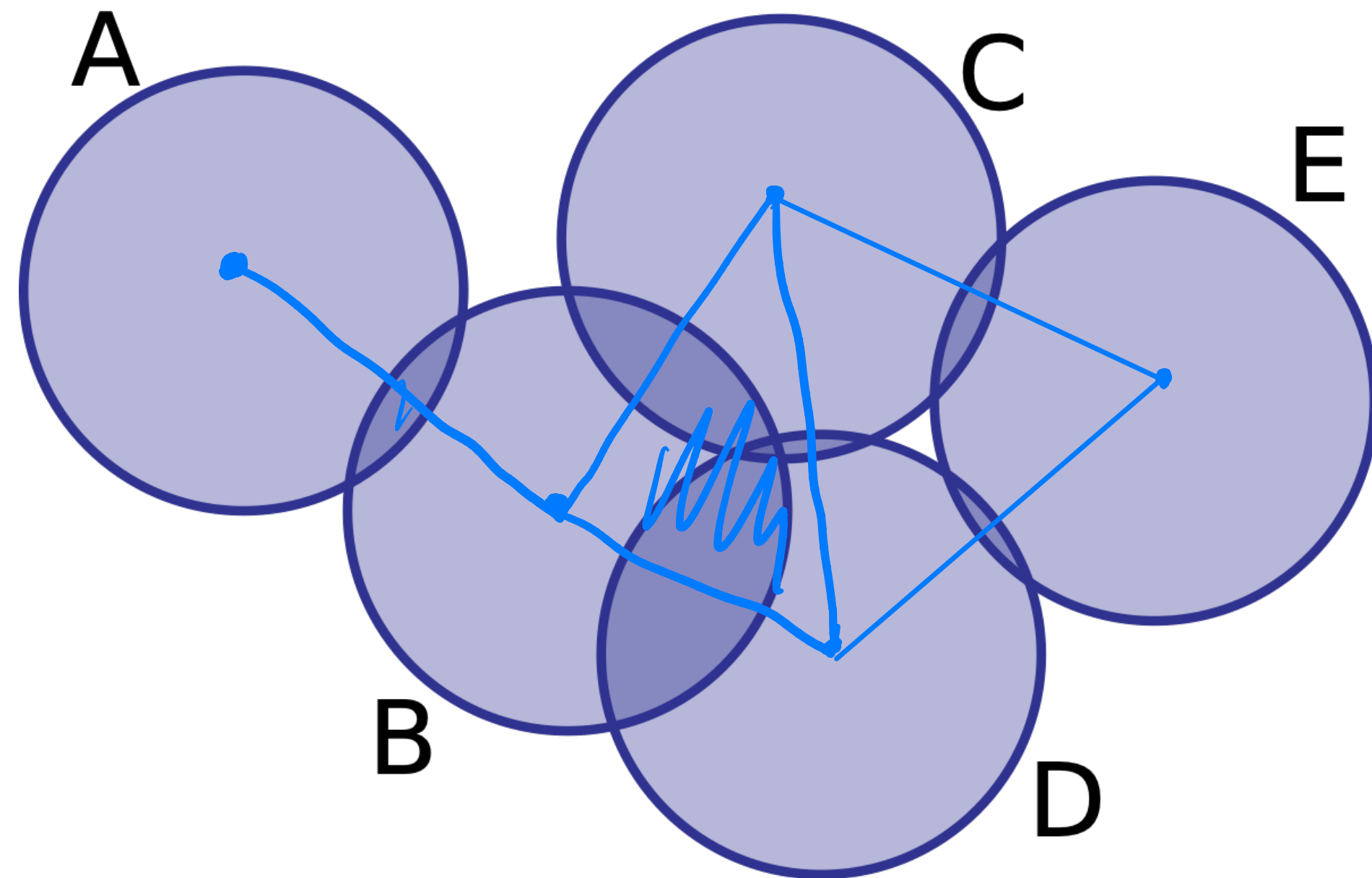
intersecting fewer sets $\Rightarrow \bigcap_{U \in X} U \subseteq \bigcap_{U \in Y} U$

$\Rightarrow Y \in \text{Nrv}(\mathcal{U})$.

Ex. if $u_1 \cap u_2 \cap u_3 \neq \emptyset$.

$\Rightarrow u_1 \cap u_2 \neq \emptyset, u_1 \cap u_3 \neq \emptyset, u_2 \cap u_3 \neq \emptyset$

Examples



Čech and Rips Complexes

Point Cloud

A point cloud is a finite collection of points in a metric space (M, d) $\mathbb{R}^d$ $\|\cdot\|_2$.

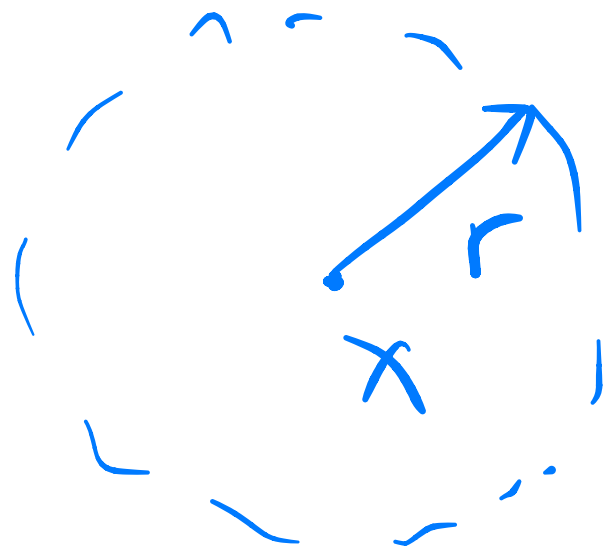
$$X = \{x_1, \dots, x_n\} \quad (\text{general position})$$

nice enough!

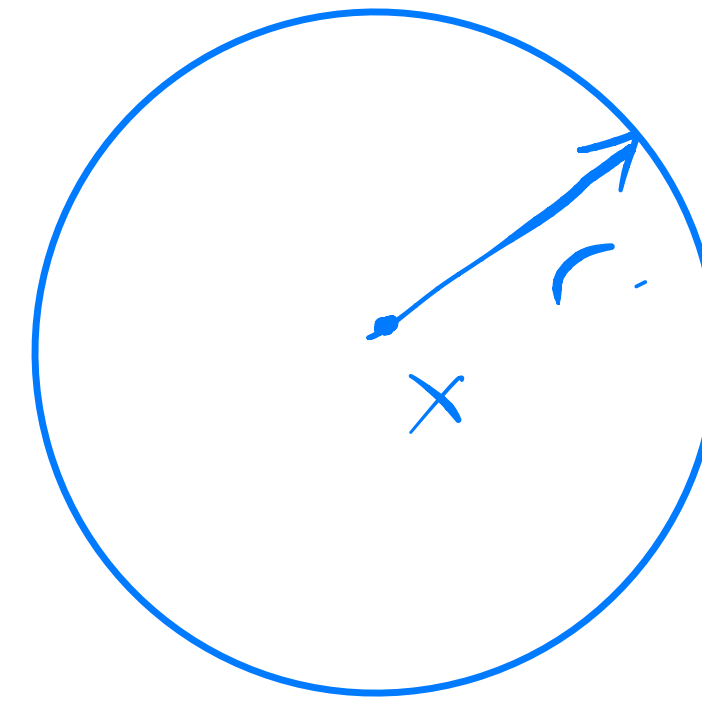


Ball

$$B_0(x, r) = \{y \in M \mid d(x, y) < r\}$$

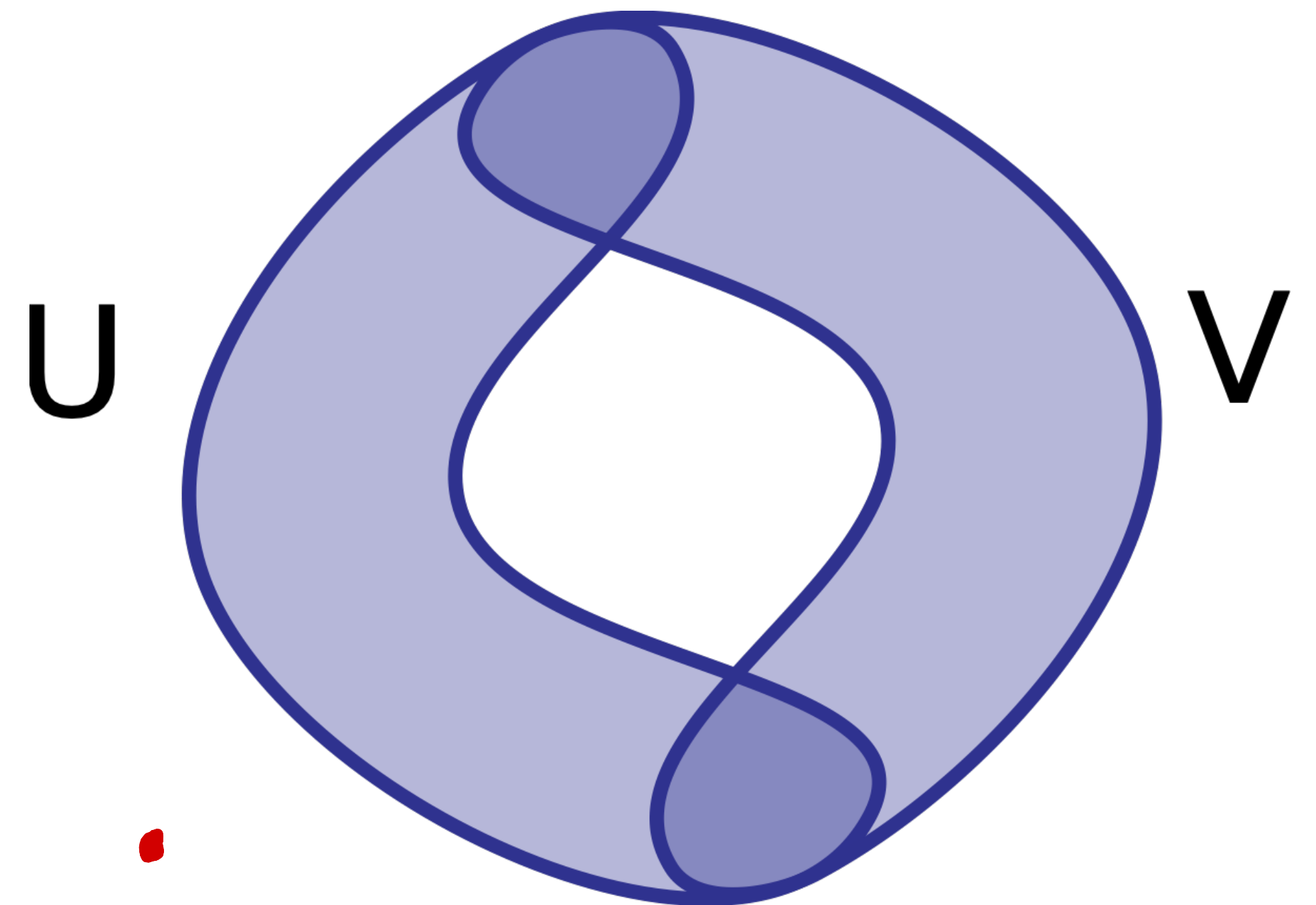
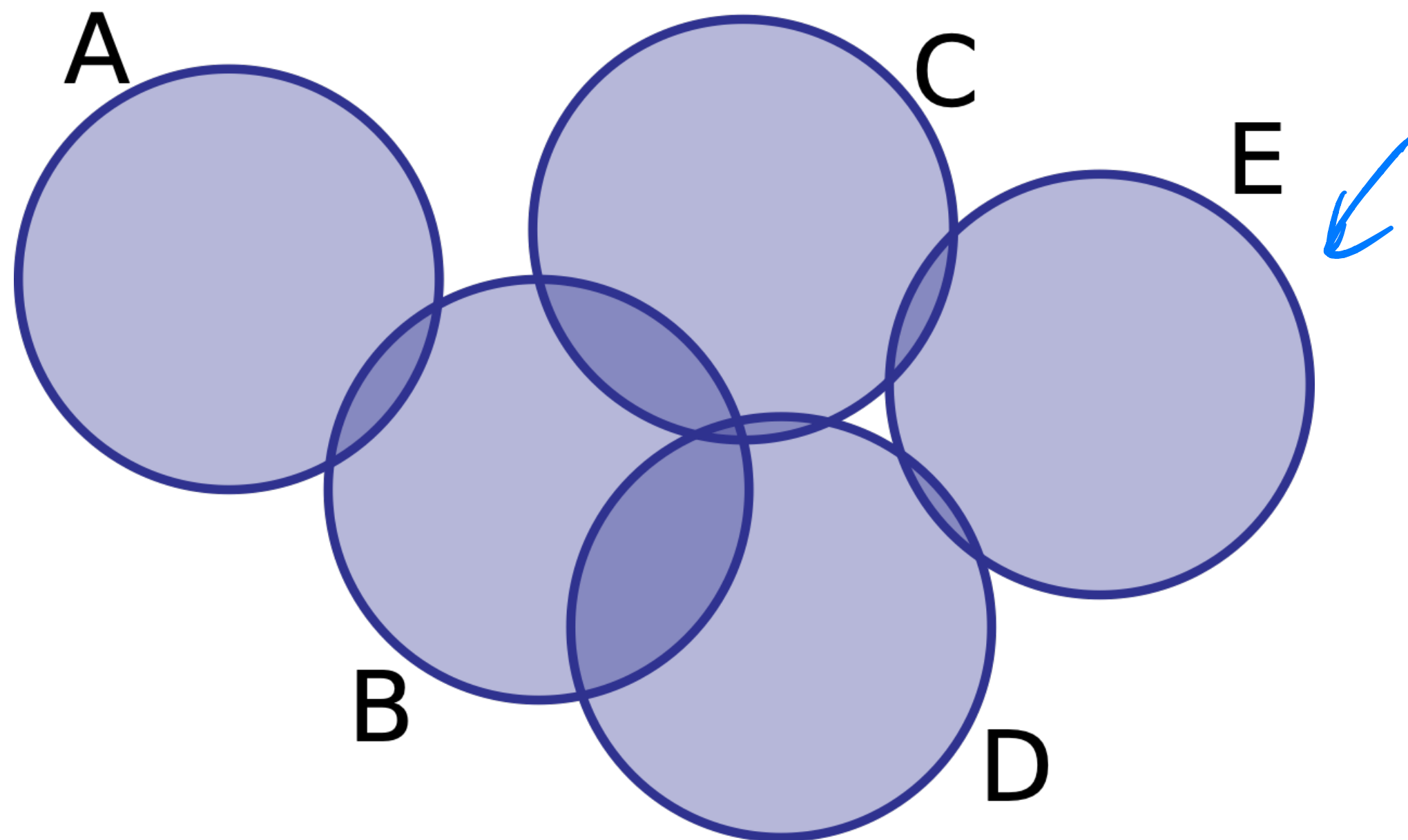


$$B(x, r) = \{y \in M \mid d(x, y) \leq r\}$$



The Nerve Lemma

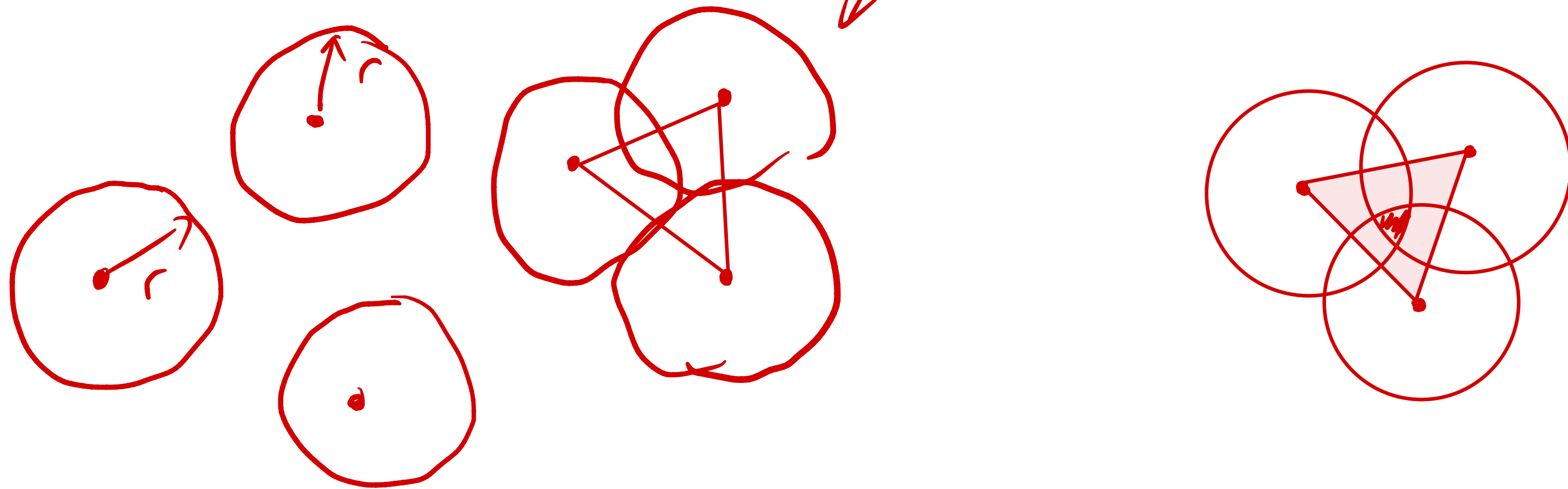
Given a finite cover $\mathcal{U}$ (open or closed) of a metric space M , the underlying space $|N(\mathcal{U})|$ is homotopy equivalent to M if every non-empty intersection $\bigcap_{i=0}^k U_{\alpha_i}$ of cover elements is homotopy equivalent to a point (contractible).



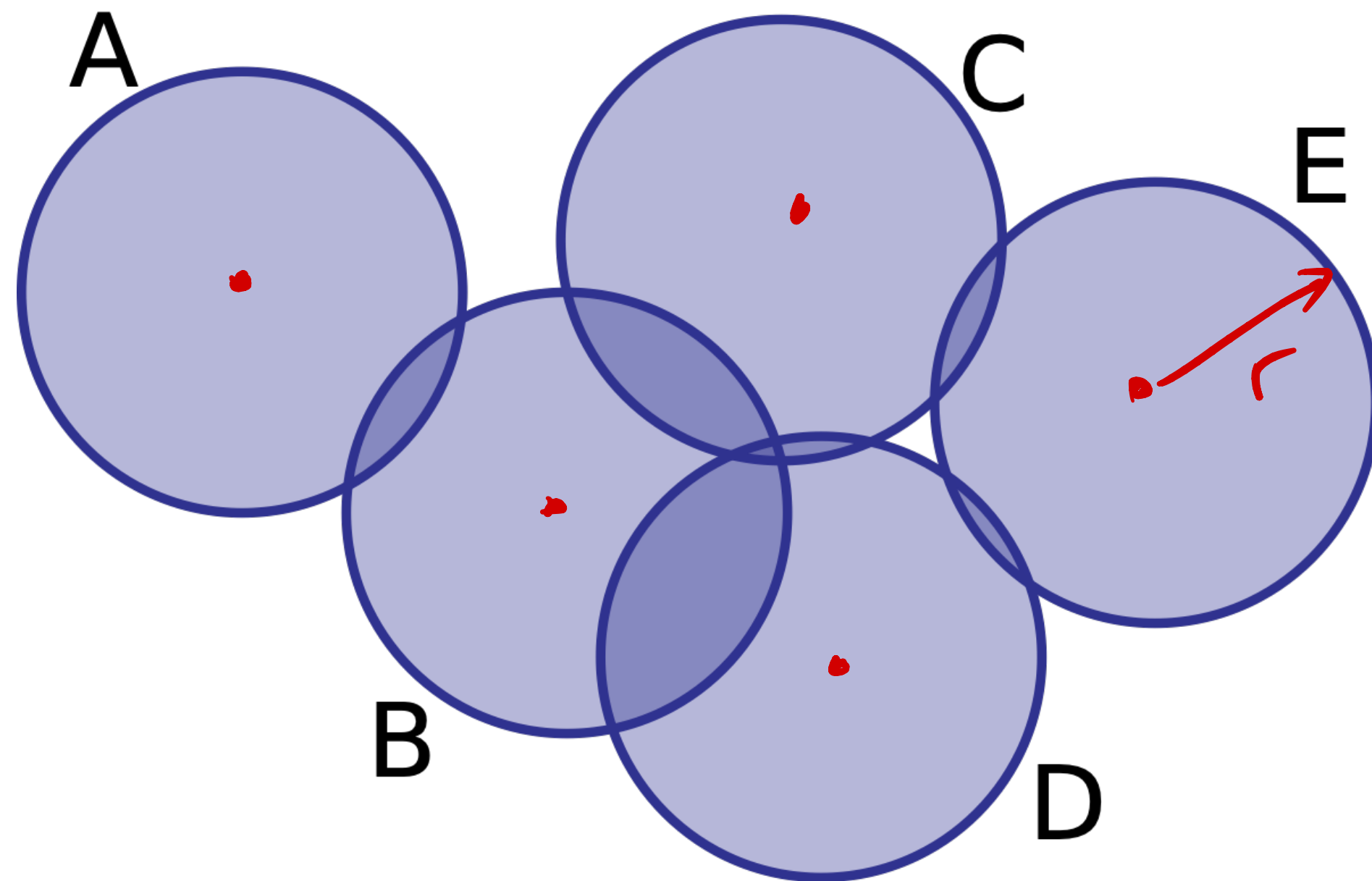
Čech Complex

Let $P \subset (M, d)$ be a finite point cloud. Fix $r \geq 0$. The Čech complex is

$$\check{C}^r(P) = \{\alpha \subseteq P \mid \bigcap_{x \in \alpha} B(x, r) \neq \emptyset\} = \text{Nrv}(\{B(x, r)\}_{x \in P})$$

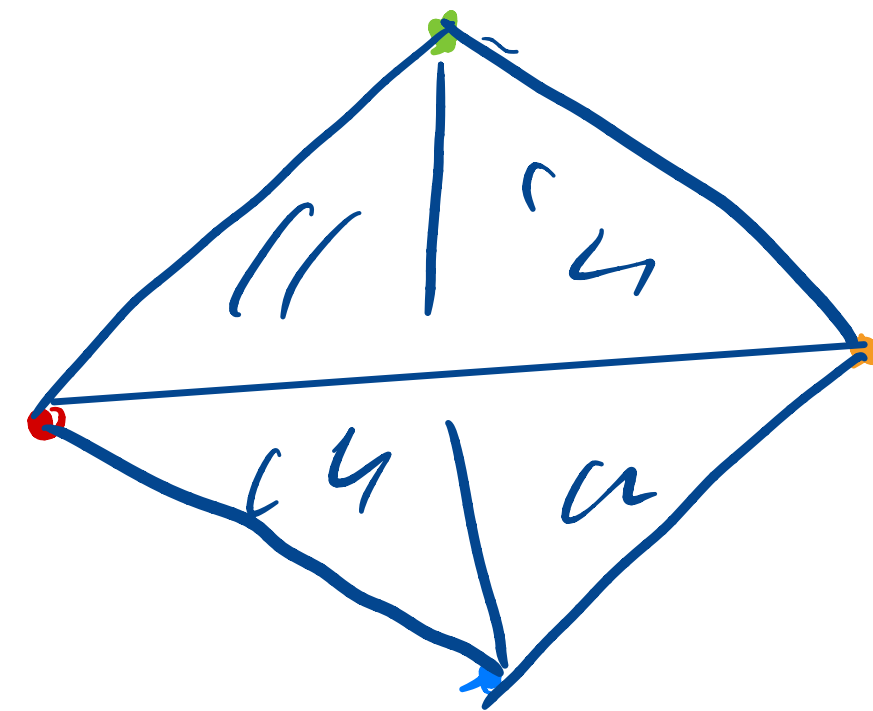
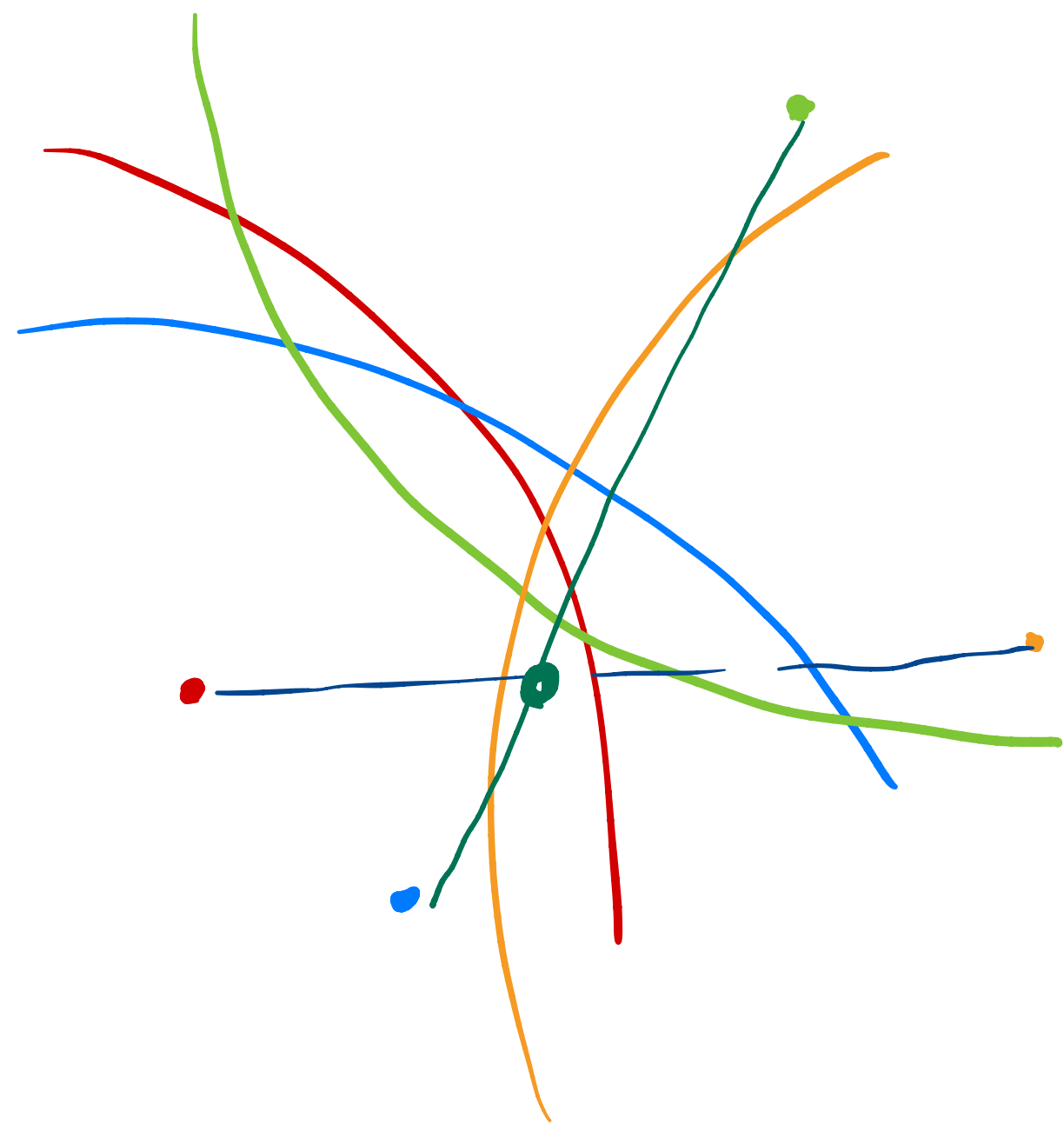


Čech Complex



Warning

The Čech complex is an **abstract** simplicial complex. The obvious map into $\mathbb{R}^d$ doesn't necessarily get you a geometric simplicial complex!

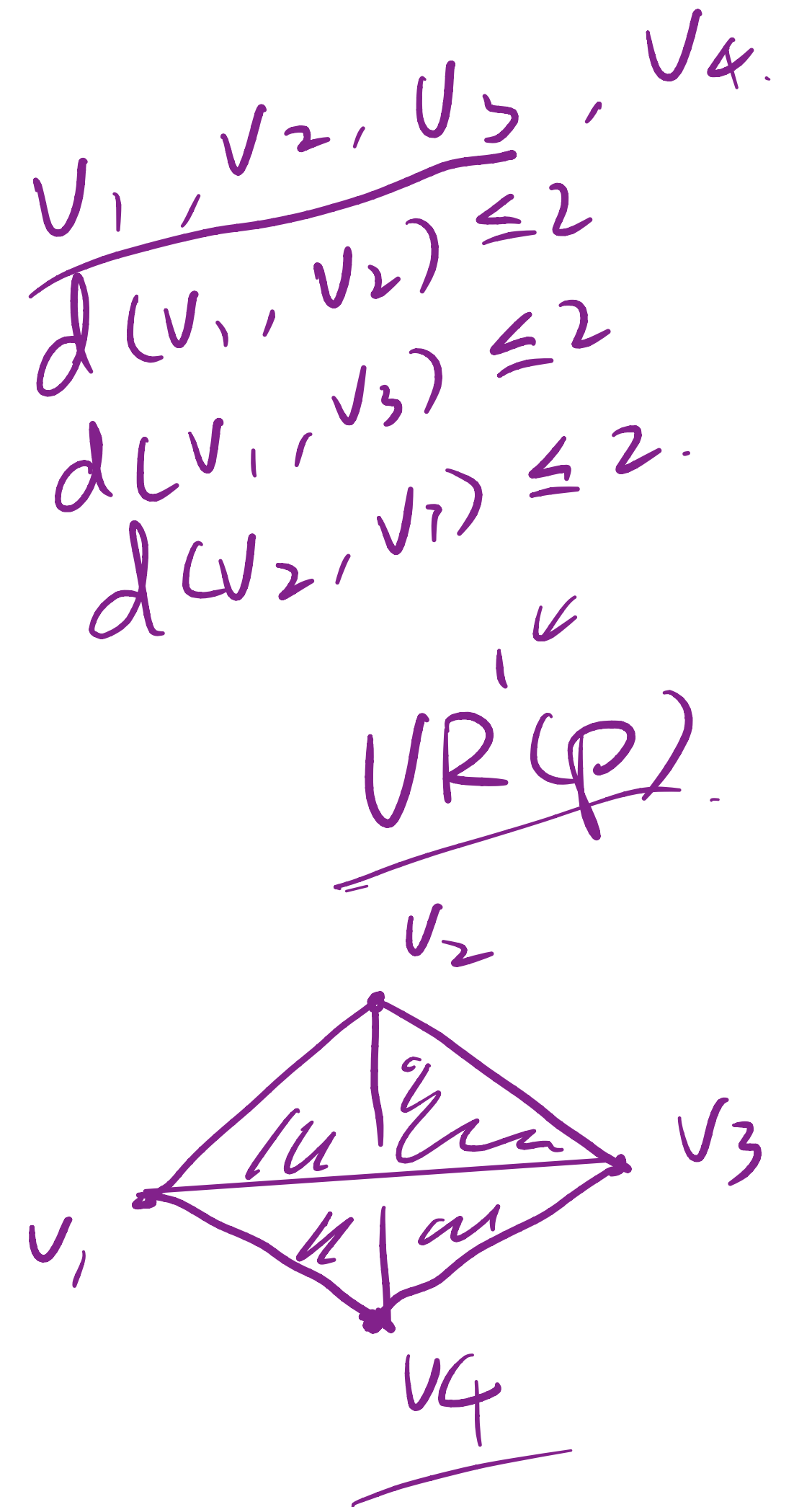
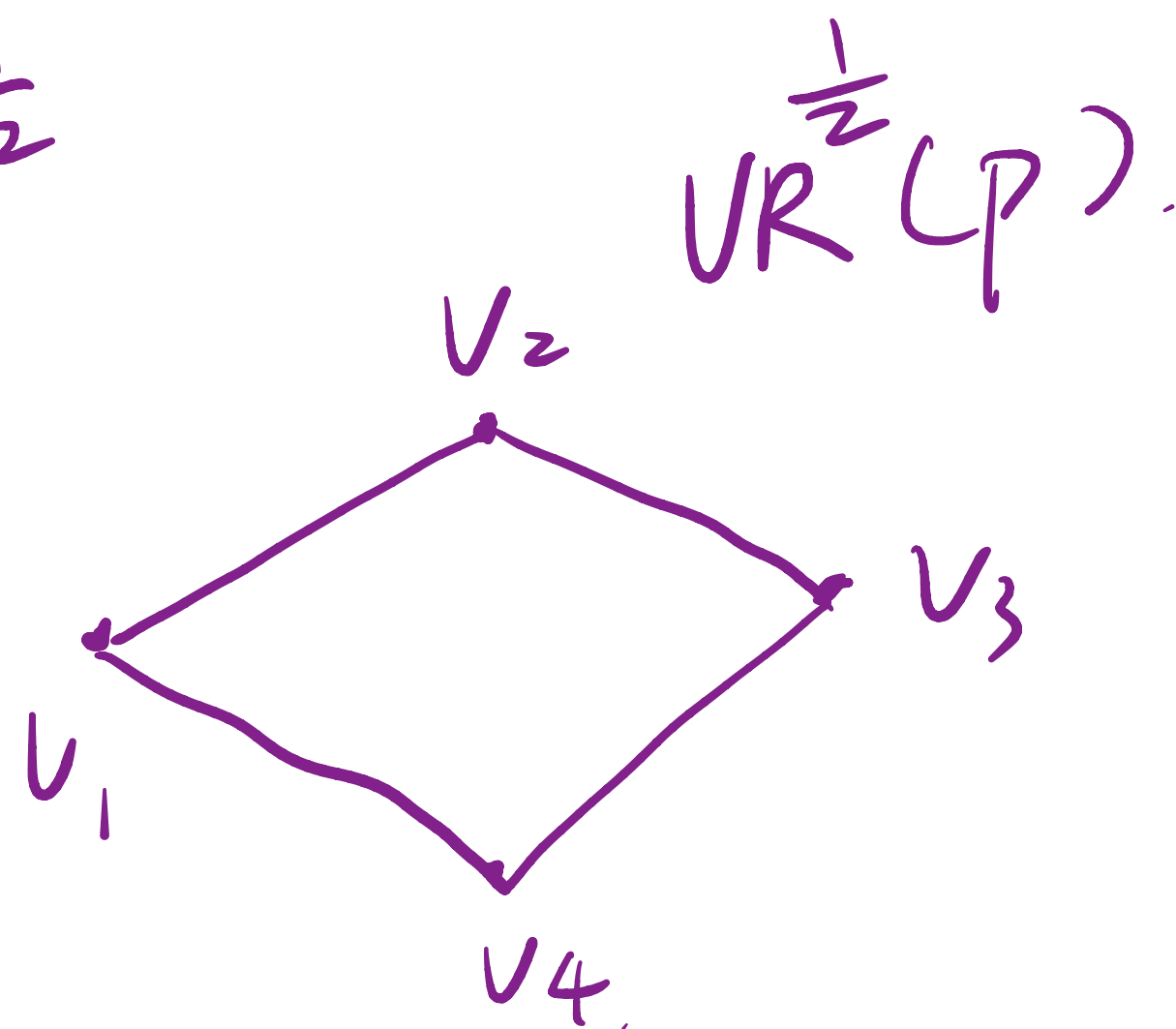
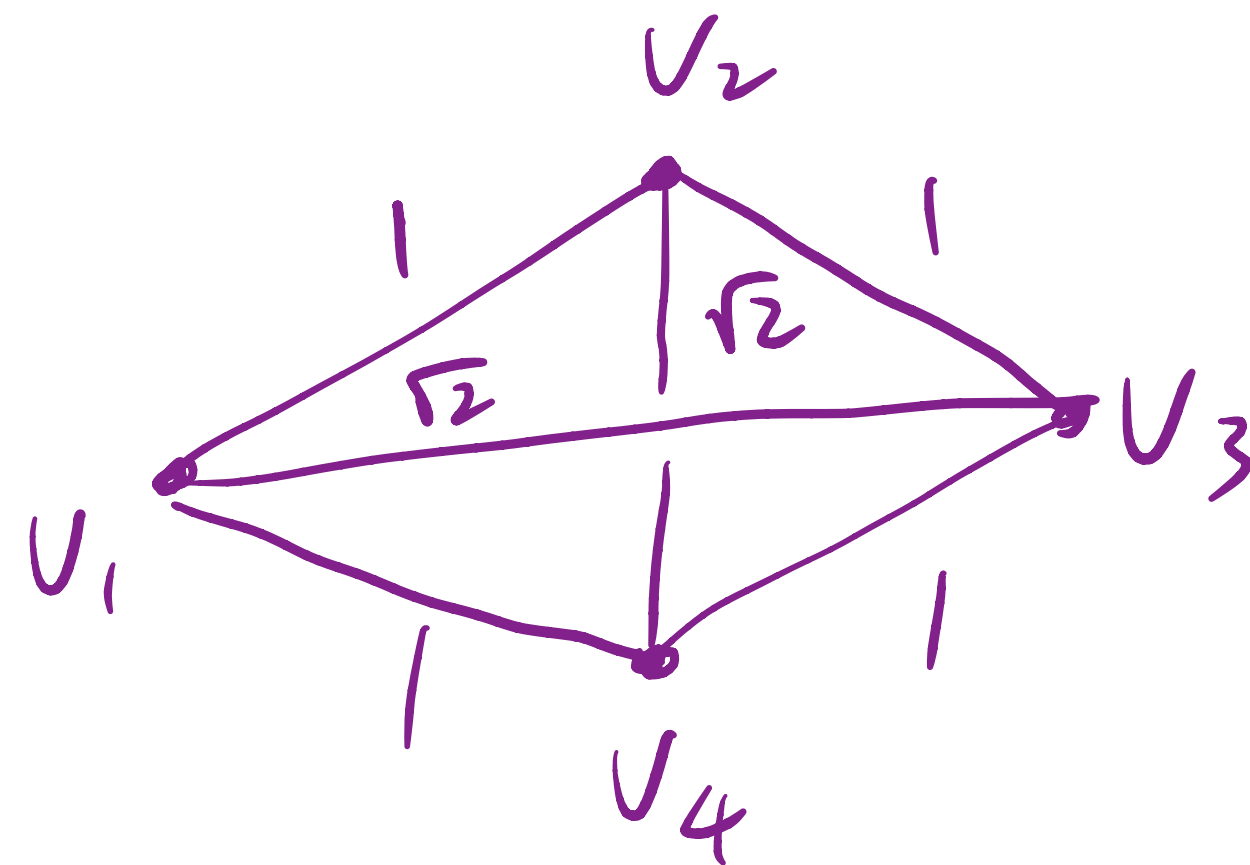


3-simplex.

Rips Complex

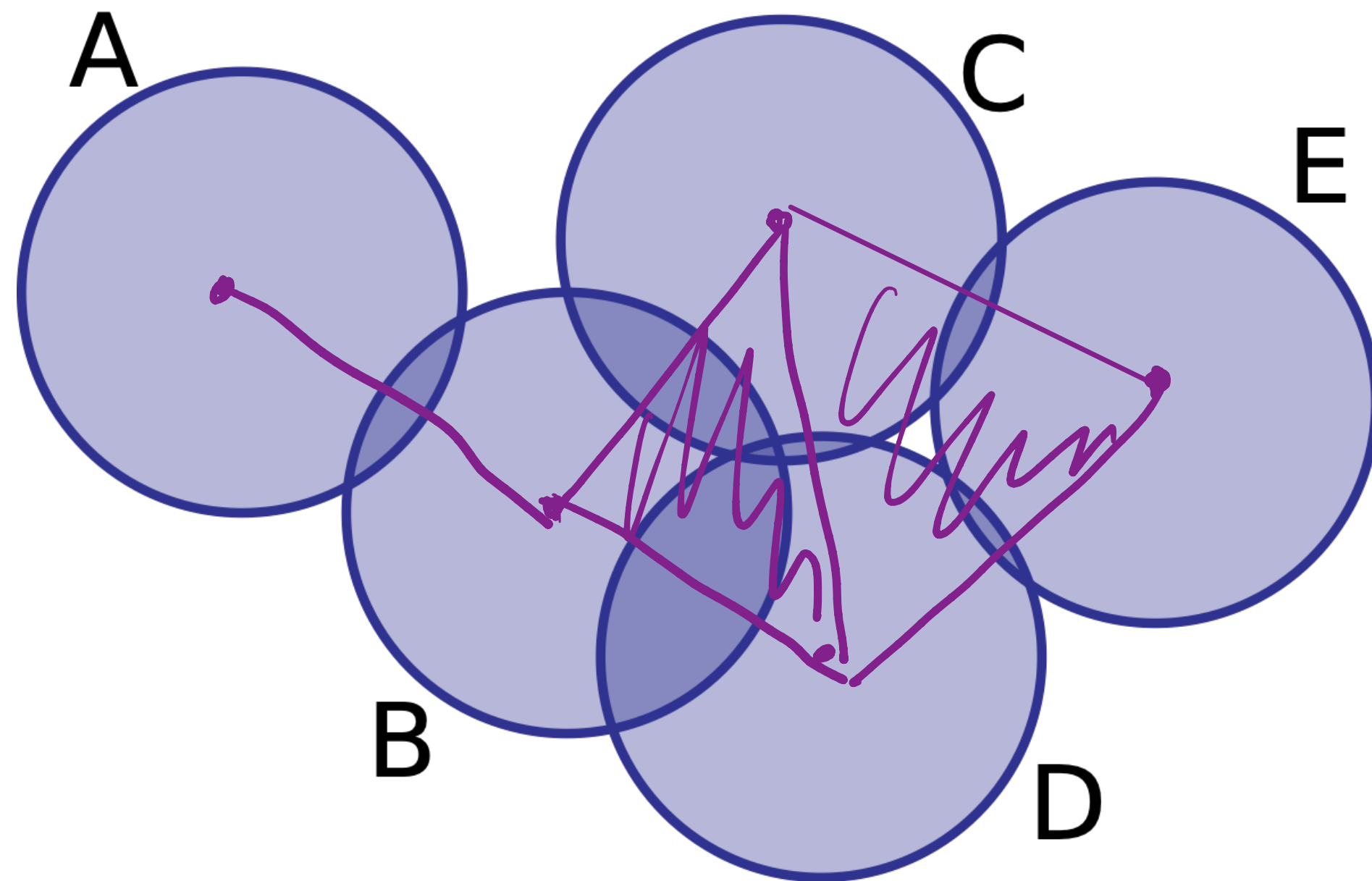
Given $P \subseteq (M, d)$, the Vietoris-Rips complex is

$$VR^r(P) = \{ \underline{\alpha} \subseteq P \mid d(\underline{x_i}, \underline{x_j}) \leq 2\underline{r} \text{ for all } \underline{x_i}, \underline{x_j} \in \alpha \}$$



$$\begin{aligned} & \underline{v_1, v_2, v_3, v_4} \\ & d(v_1, v_2) \leq 2 \\ & d(v_1, v_3) \leq 2 \\ & d(v_2, v_3) \leq 2 \end{aligned}$$

Rips Complex



CE
 ED
 CD

Rips-Čech Lemma

Given point cloud $P \subset (M, d)$ and $r > 0$, $\check{C}^r(P) \subseteq \text{VR}^r(P) \subseteq \check{C}^{2r}(P)$

more computable



topologically
accurate.

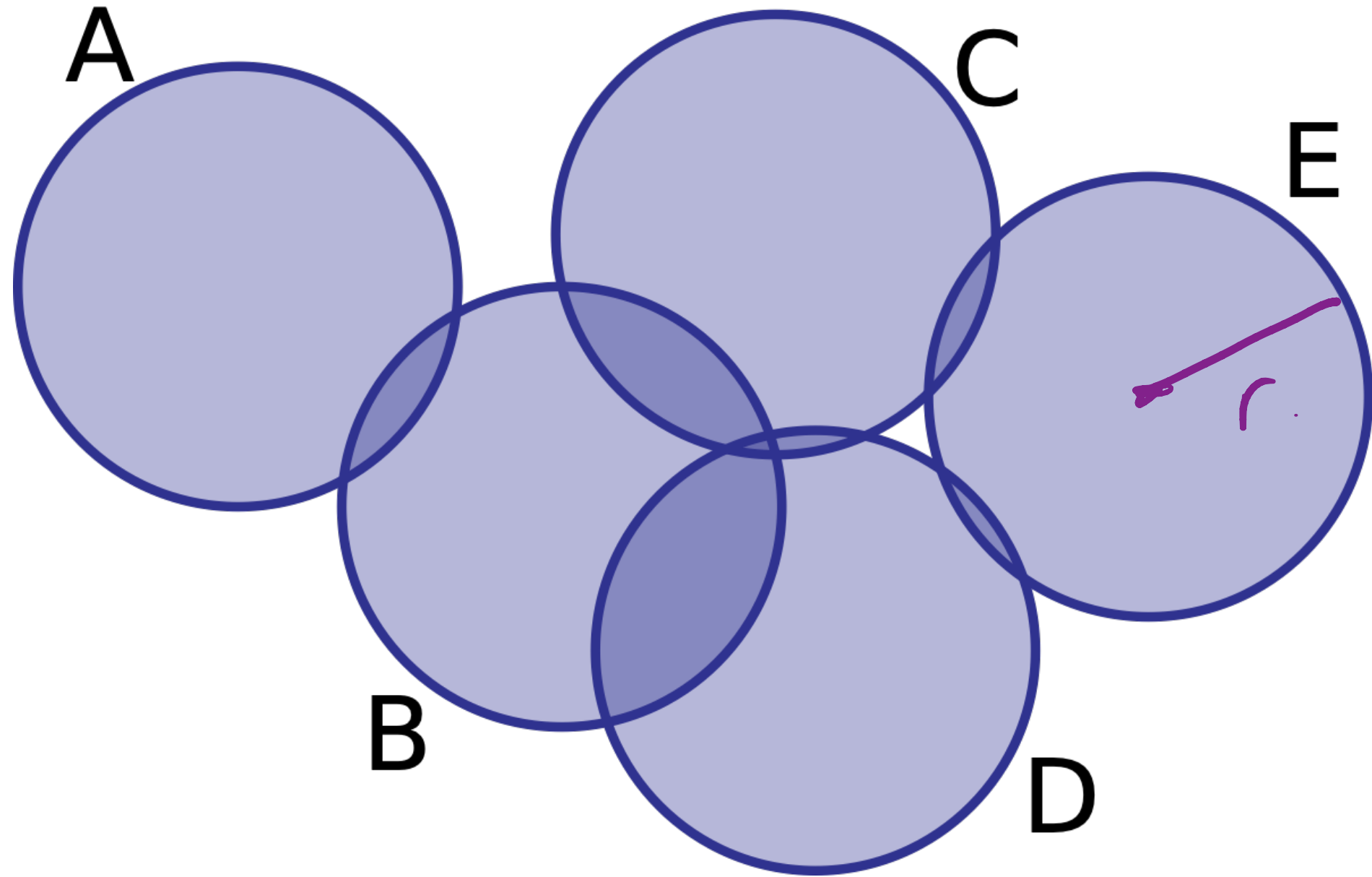
$\{x_1, \dots, x_5\}$

4-simplex?

$\binom{5}{2} = 10$ pairwise relations.

Warning: Radius vs Diameter

$$VR^r(P) = \{\alpha \subseteq P \mid d(x_i, x_j) \leq \underline{2r} \text{ for all } x_i, x_j \in \alpha\}$$



$$VR^{\text{diam}} = \{ \dots \mid d(\cdot, \cdot) \leq \text{diam} \dots \}$$

Alpha Complex

Rips.



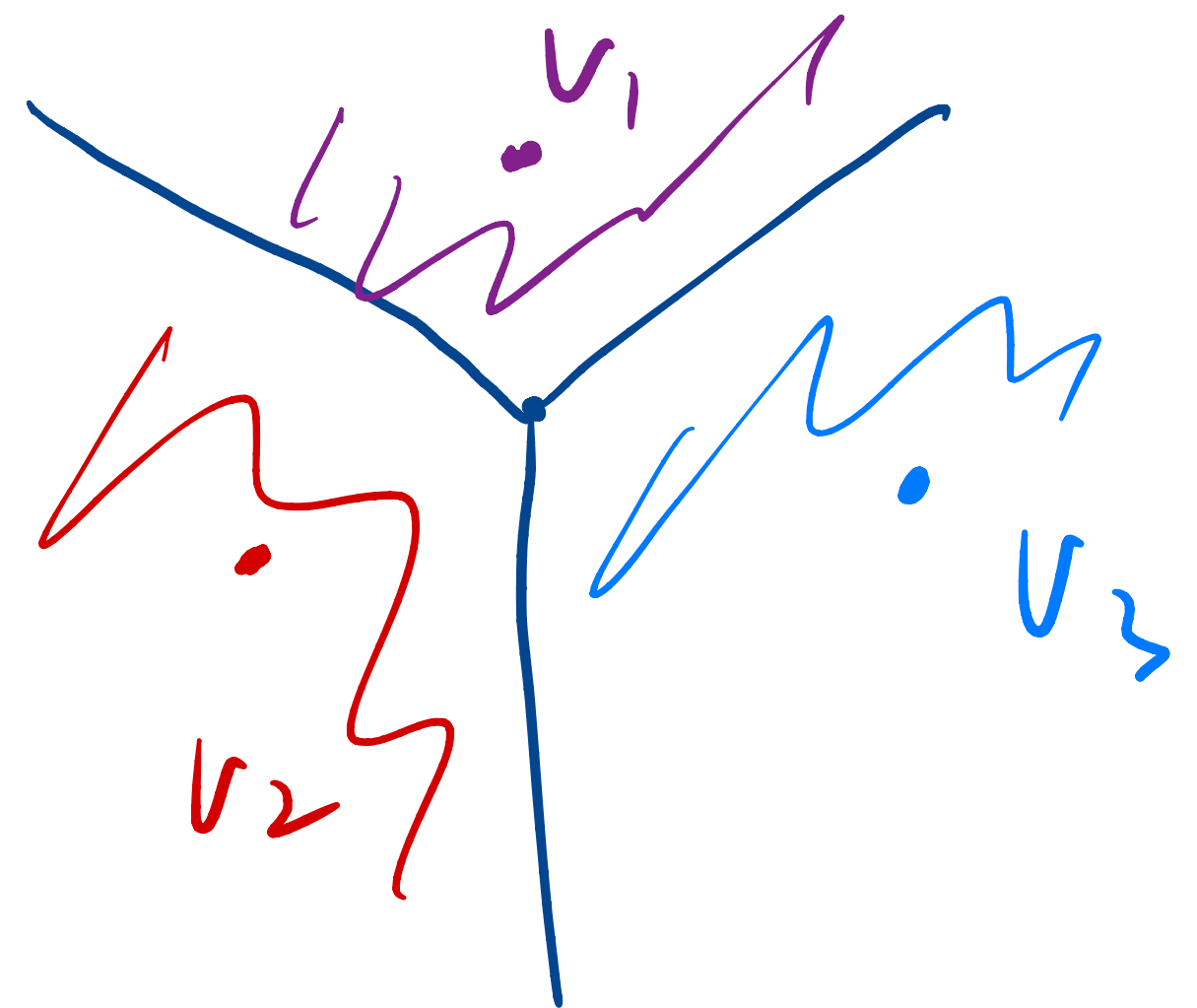
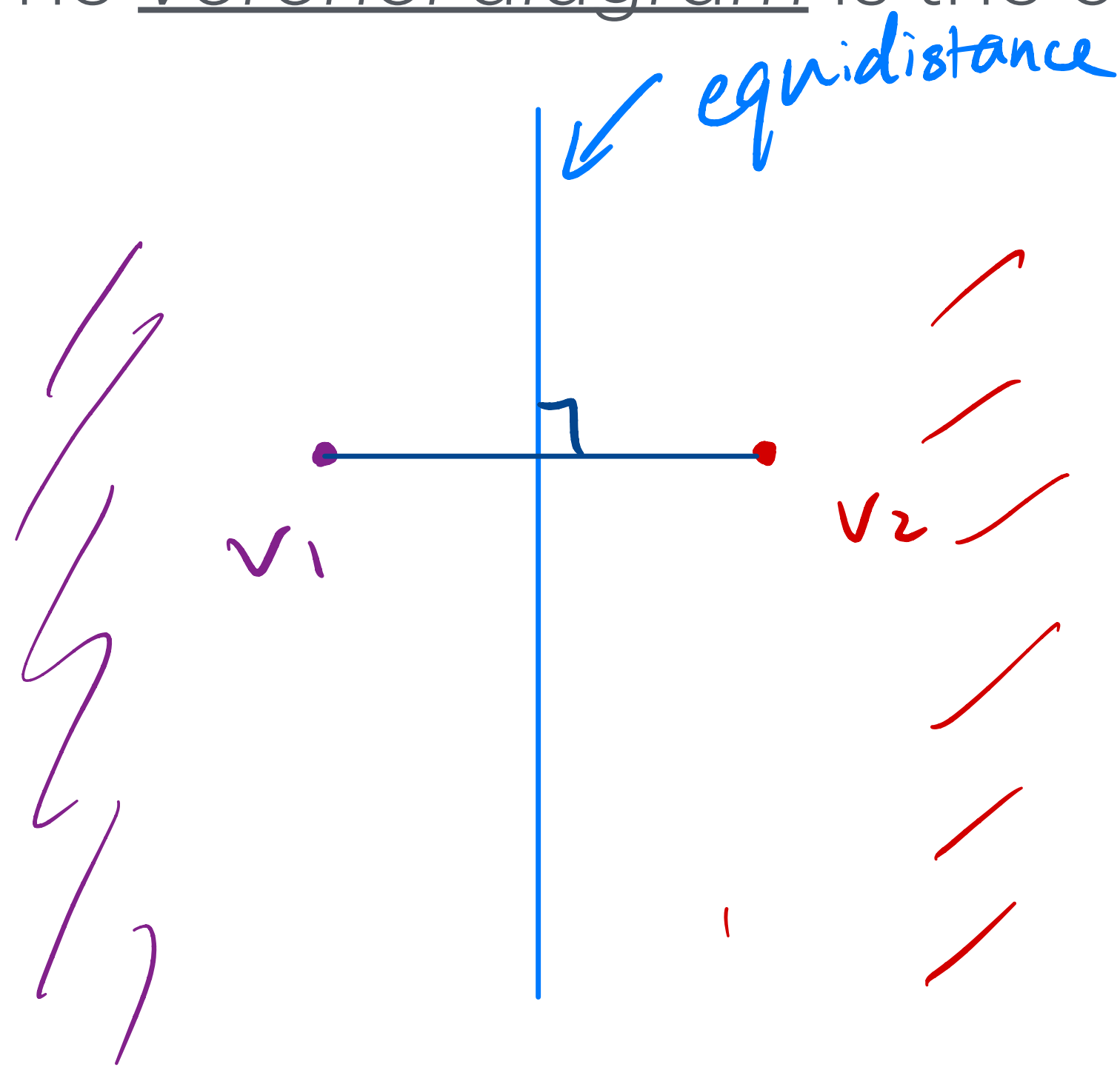
of simplices grows exponentially.

Voronoi Diagram

Given point cloud $P \subseteq \mathbb{R}^d$. The Voronoi cell of $u \in P$ is

$$V_u = \{x \in \mathbb{R}^d \mid \|x - u\| \leq \|x - v\|, v \in P\}$$

The Voronoi diagram is the collection of Voronoi cells $\text{Vor}(P) = \{V_u \mid u \in P\}$



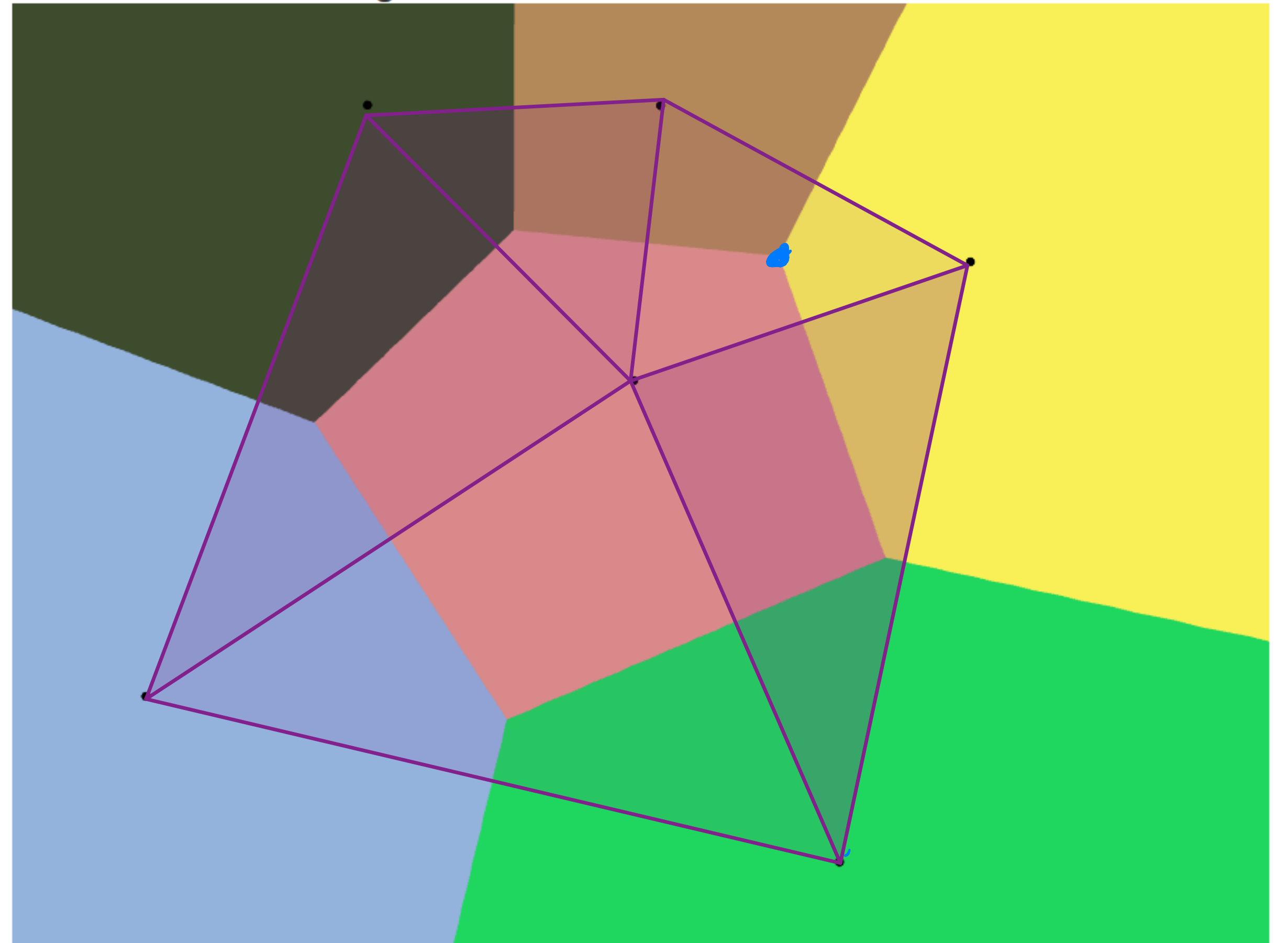
Delaunay Triangulation

The Delaunay complex of point cloud $P \subseteq \mathbb{R}^d$ is the nerve of the Voronoi Diagram

$$\text{Del}(P) = \{\sigma \subseteq P \mid \bigcap_{u \in \sigma} V_u \neq \emptyset\}$$

$\mathbb{R}^2$ $\downarrow$ 2-simplex
 $\mathbb{R}^3$ $\downarrow$ 3-simplex

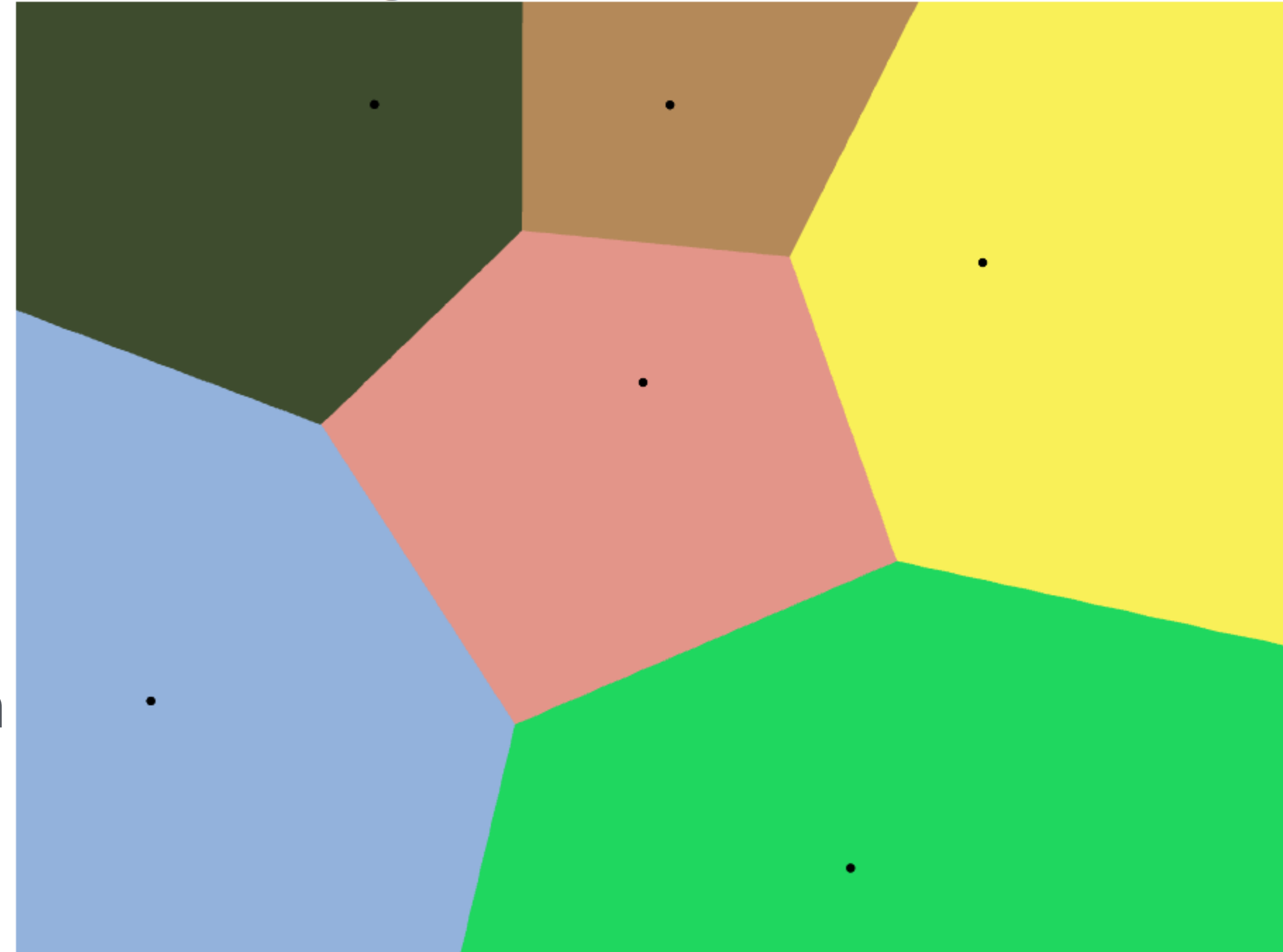
Interactive Voronoi Diagram Generator with WebGL



Properties

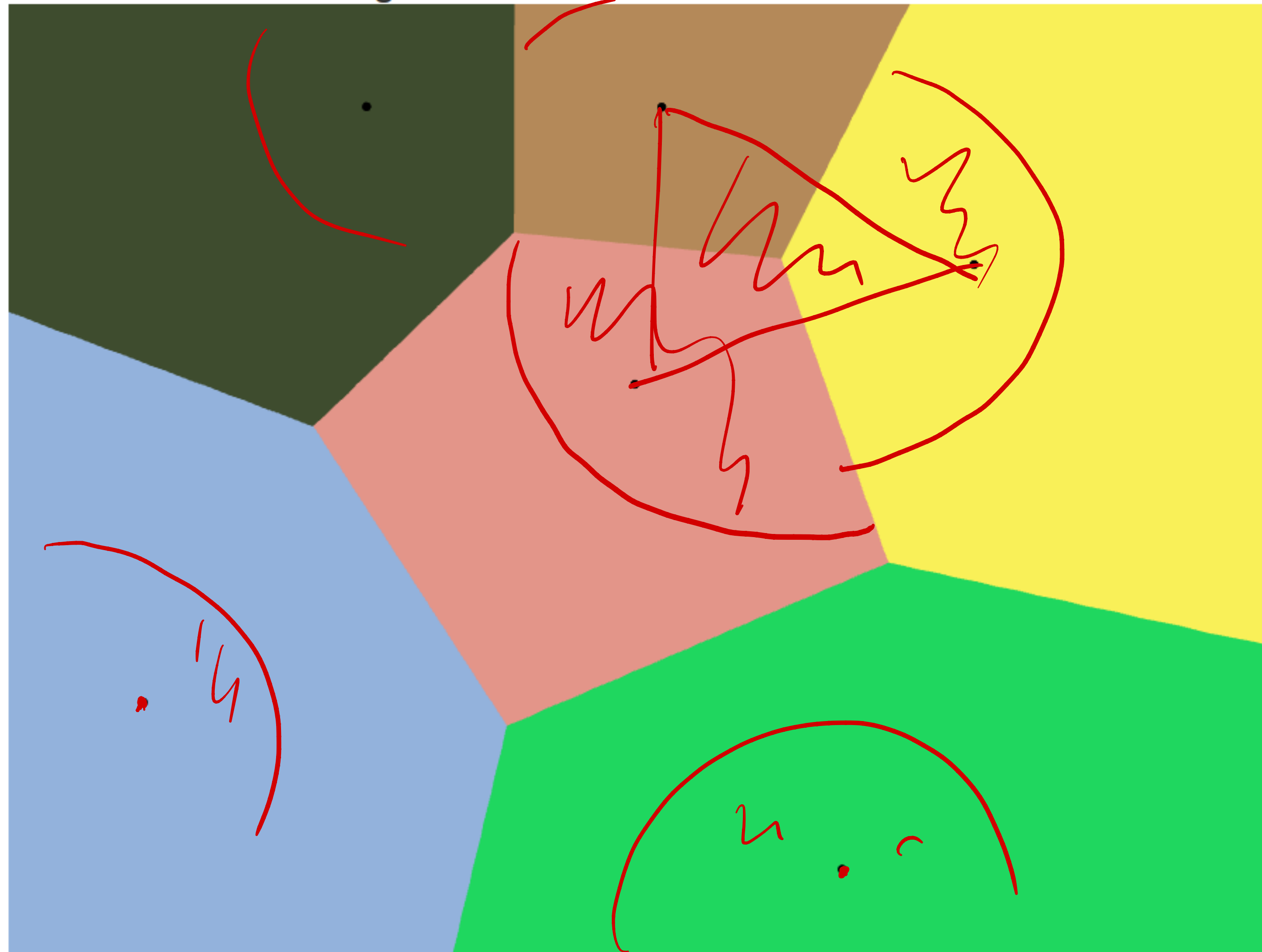
- Delauney is an abstract simplicial complex
- If we have points in general position, the obvious embedding gives a geometric simplicial complex *nice enough.*
- Delauney is FIXED (has nothing to do with a radius or diameter parameter)

Interactive Voronoi Diagram Generator with WebGL



Alpha Complex

Interactive Voronoi Diagram Generator with WebGL



Alpha Complex

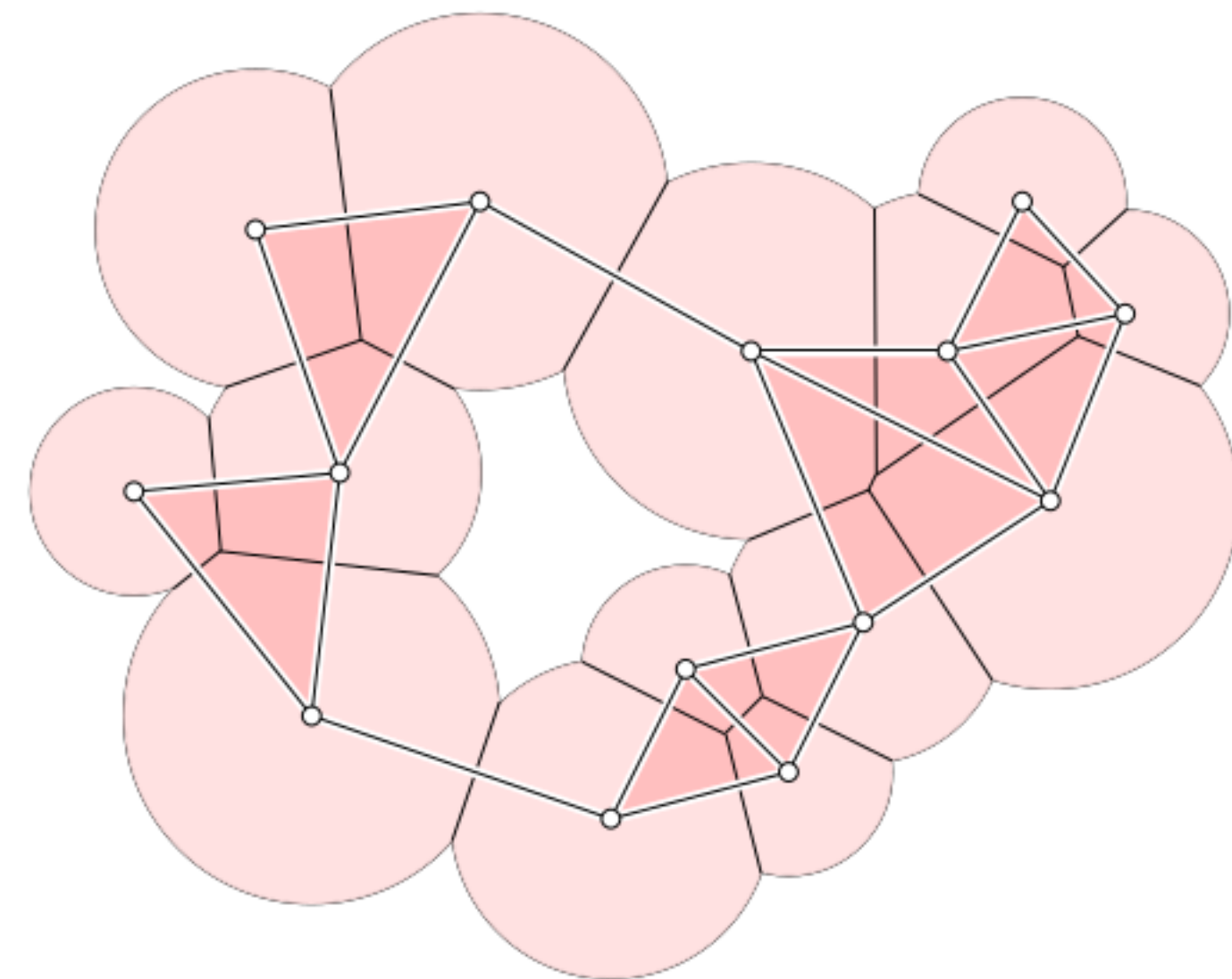
The restricted Voronoi cell at scale r is

$$V_p^r = V_P \cap B(p, r) = \{x \in B(p, r) : d(x, p) \leq d(x, q) \text{ for all } q \in P\}$$

The alpha complex for point cloud P with radius $r \geq 0$ is the nerve

$$\text{Alpha}^r(P) = \text{Nrv}(\{V_p^r \mid p \in P\}).$$

$$\text{Equivalently } \text{Alpha}^r(P) = \{\sigma \subseteq P : \bigcap_{p \in \sigma} (V_p \cap B(p, r)) \neq \emptyset\}$$



Properties

- $\text{Alpha}(r) \subseteq \text{Delaunay}$

- $\text{Alpha}(r) \subseteq \check{C}(r)$

- $\text{Alpha}(r)$ has the same homotopy type as the union of balls of radius r

topological guarantee

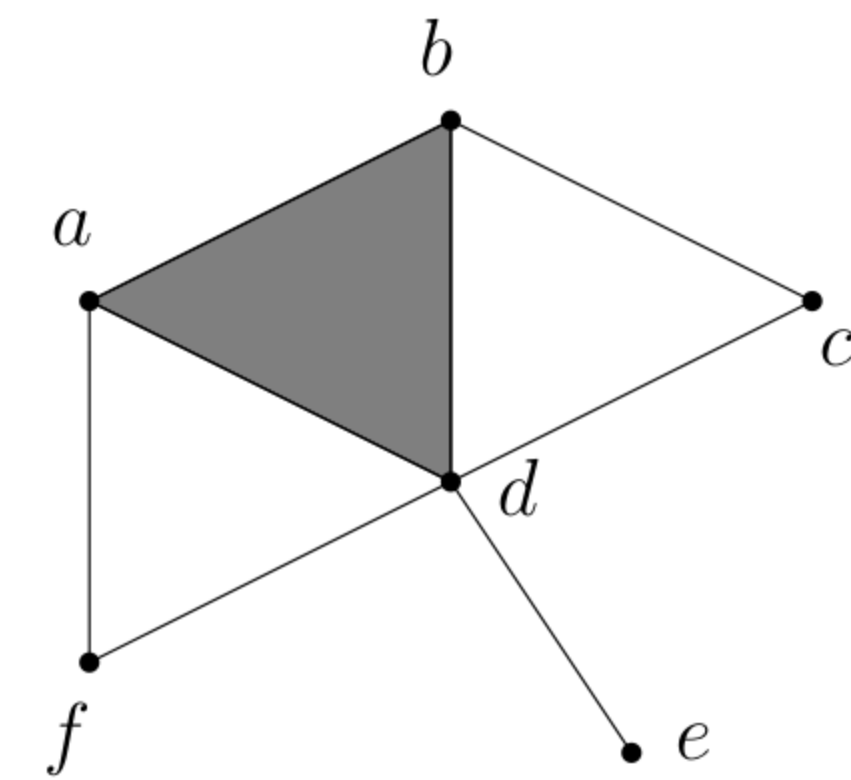
homotopic equivalent.

Problems to present (choose one)

- A flag in a simplicial complex K is a nested sequence of proper faces $\sigma_0 < \sigma_1 < \dots < \sigma_k$.

Show that the collection of flags in K forms a simplicial complex; this is called the order complex of K . Show the construction on the example below.

Isabelle



Jason.

- Prove the Rips-Čech Lemma and show that the factor of 2 cannot be improved for arbitrary metric spaces.